



US012405341B2

(12) **United States Patent**
Kang et al.

(10) **Patent No.:** **US 12,405,341 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **DIRECTIONAL ACOUSTIC SENSOR**

(56) **References Cited**

(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(72) Inventors: **Sungchan Kang**, Hwaseong-si (KR);
Cheheung Kim, Yongin-si (KR);
Choongho Rhee, Anyang-si (KR)

(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 126 days.

(21) Appl. No.: **17/731,742**

(22) Filed: **Apr. 28, 2022**

(65) **Prior Publication Data**

US 2023/0152411 A1 May 18, 2023

(30) **Foreign Application Priority Data**

Nov. 12, 2021 (KR) 10-2021-0155987

(51) **Int. Cl.**
G01S 3/801 (2006.01)
G01S 3/803 (2006.01)
H03H 9/13 (2006.01)

(52) **U.S. Cl.**
CPC **G01S 3/801** (2013.01); **G01S 3/803** (2013.01); **H03H 9/132** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

U.S. PATENT DOCUMENTS

3,886,504 A * 5/1975 Hartmann H03H 9/0038 333/195
4,885,781 A * 12/1989 Seidel G01H 11/06 381/174

(Continued)

FOREIGN PATENT DOCUMENTS

CN 212163296 U * 12/2020
CN 113507676 A 10/2021
(Continued)

OTHER PUBLICATIONS

Brown, Joseph J., et al. "Nonlinear Mechanics of Interlocking Cantilevers." Journal of Applied Mechanics 84.12 (2017): 121012. (Year: 2017).*

(Continued)

Primary Examiner — Isam A Alsomiri

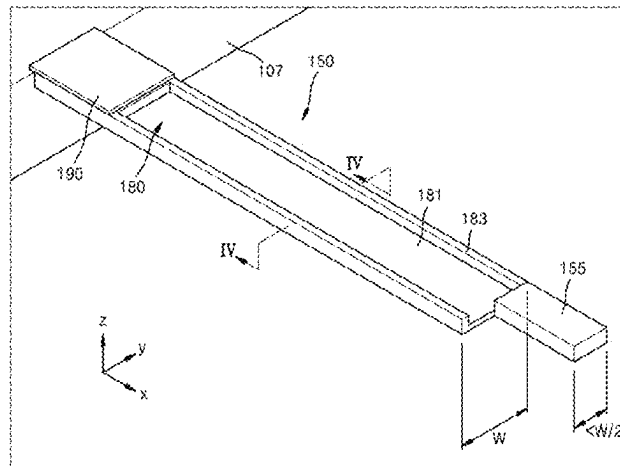
Assistant Examiner — Jonathan D Armstrong

(74) *Attorney, Agent, or Firm* — Sughrue Mion, PLLC

(57) **ABSTRACT**

A directional acoustic sensor includes: a support including a first support portion and a second support portion that are separated from each other and face each other; a plurality of first resonators extending in a length direction thereof from the first support portion of the support; and a plurality of second resonators extending in the length direction thereof from the second support portion of the support and facing the plurality of first resonators, wherein each first resonator of the plurality of first resonators has a first end, wherein each second resonator of the plurality of second resonators has a second end, and wherein, in a first resonator arrangement of a region where the plurality of first resonators and the plurality of second resonators face each other, the first ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.

23 Claims, 25 Drawing Sheets



(56)

References Cited**U.S. PATENT DOCUMENTS**

5,491,604 A * 2/1996 Nguyen H03H 9/02425
333/197

5,856,722 A 1/1999 Haronian et al.

5,955,932 A * 9/1999 Nguyen G01R 33/0286
361/278

6,012,334 A * 1/2000 Ando G01H 3/08
73/648

6,079,274 A * 6/2000 Ando G01H 11/08
73/649

6,236,281 B1 * 5/2001 Nguyen H03H 9/02393
361/278

6,438,243 B1 * 8/2002 Ikeuchi H04R 15/02
381/174

6,515,401 B1 2/2003 Ando et al.

6,651,504 B1 * 11/2003 Datskos G01H 11/06
73/632

8,008,105 B2 * 8/2011 Huang B06B 1/0292
438/52

8,103,027 B2 1/2012 Zhang et al.

8,170,244 B2 5/2012 Ryan et al.

9,148,729 B2 9/2015 Josefsson

10,141,007 B1 11/2018 Kim et al.

10,243,535 B2 * 3/2019 Kawasaki H03H 9/1092

10,356,509 B2 * 7/2019 Park H04R 17/025

10,541,670 B2 * 1/2020 Yoon H03H 3/0076

11,146,240 B2 * 10/2021 Yoon H03H 9/2426

11,359,960 B2 * 6/2022 Kang H04R 17/10

2005/0005676 A1 * 1/2005 Crawley B81B 3/0089
73/24.01

2011/0140579 A1 * 6/2011 Moon H02N 2/186
29/25.35

2011/0215680 A1 9/2011 Yamada et al.

2012/0062340 A1 * 3/2012 Huang H03H 9/462
333/186

2013/0051588 A1 * 2/2013 Ruile H03H 9/1457
29/25.35

2013/0135264 A1 * 5/2013 Black H03H 9/178
216/13

2014/0009032 A1 * 1/2014 Takahashi H03H 9/02559
29/25.35

2015/0263700 A1 * 9/2015 Yasumoto H01L 28/20
310/313 B

2016/0050506 A1 * 2/2016 Kim H04R 1/245
381/56

2017/0006385 A1 1/2017 Kim

2017/0179920 A1 * 6/2017 Kawasaki H03H 9/725

2018/0097506 A1 4/2018 Kang et al.

2018/0130485 A1 5/2018 Park et al.

2018/0131347 A1 * 5/2018 Rhee H03H 9/02259

2018/0138886 A1 5/2018 Yoon et al.

2019/0033904 A1 1/2019 Hong et al.

2019/0072635 A1 3/2019 Kang et al.

2019/0131954 A1 * 5/2019 Okada H03H 9/02992

2019/0174244 A1 6/2019 Kim et al.

2019/0200119 A1 6/2019 Kang et al.

2019/0348050 A1 11/2019 Kim et al.

2020/0204187 A1 6/2020 Hong et al.

2020/0212885 A1 7/2020 Kim et al.

2022/0014172 A1 1/2022 Nozoe

2023/0270010 A1 * 8/2023 Luo F16K 31/006
251/129.06

FOREIGN PATENT DOCUMENTS

JP 3729054 B2 12/2005

JP 6856820 B2 4/2021

KR 102207928 B1 1/2021

OTHER PUBLICATIONS

Communication dated Nov. 8, 2022 issued by the European Patent Office in European Application No. 22173724.0.

Kang, S., et al., "Directional Sound Sensor With Consistent Directivity and Sensitivity in the Audible Range", Journal of Microelectromechanical Systems, vol. 30, No. 3, Jun. 2021, pp. 471-479.

* cited by examiner

FIG. 1

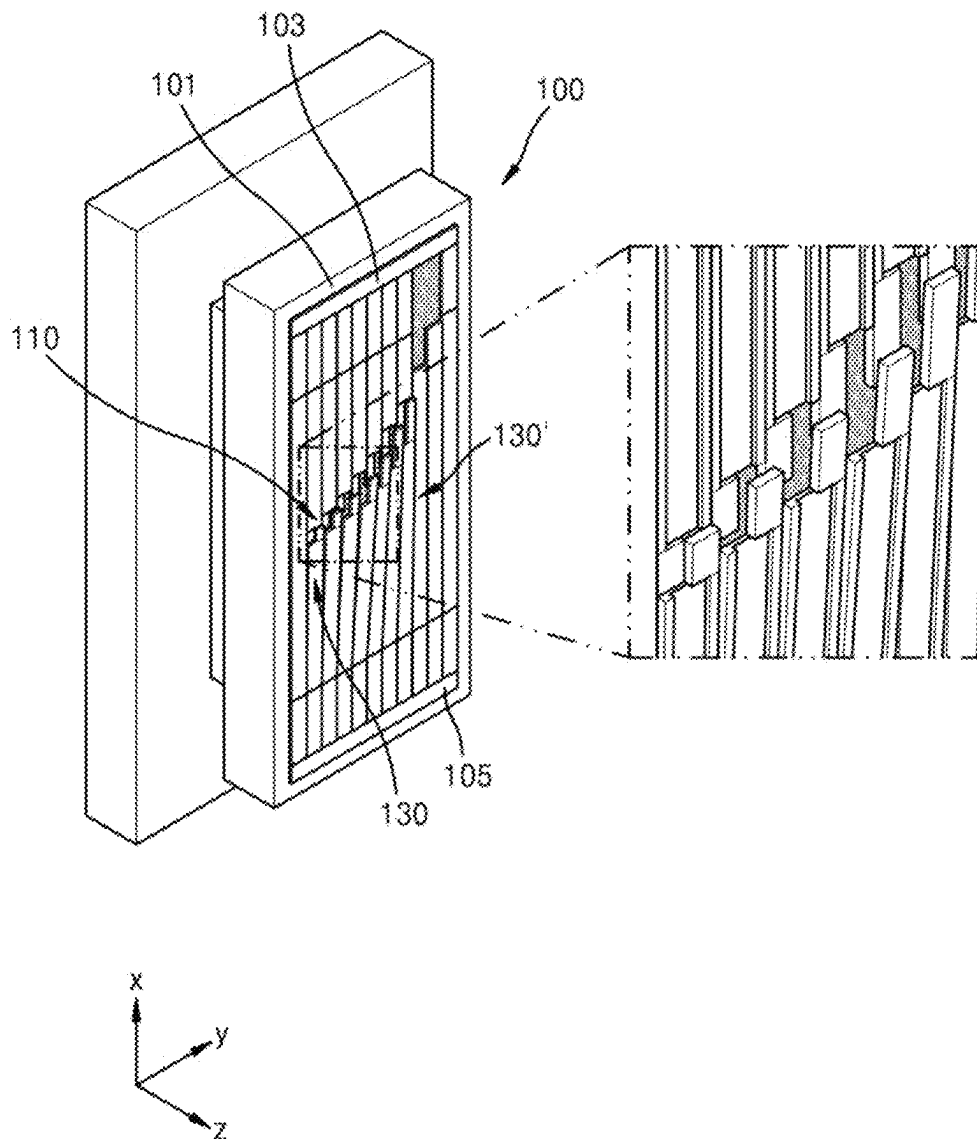


FIG. 2

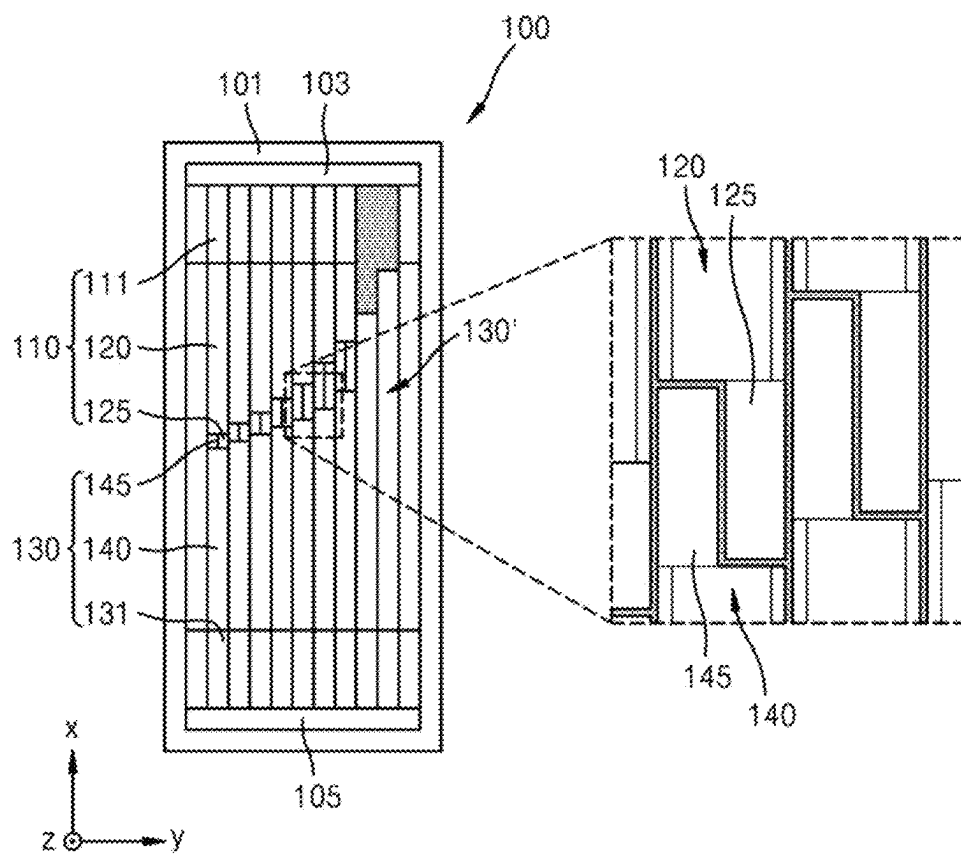


FIG. 3A

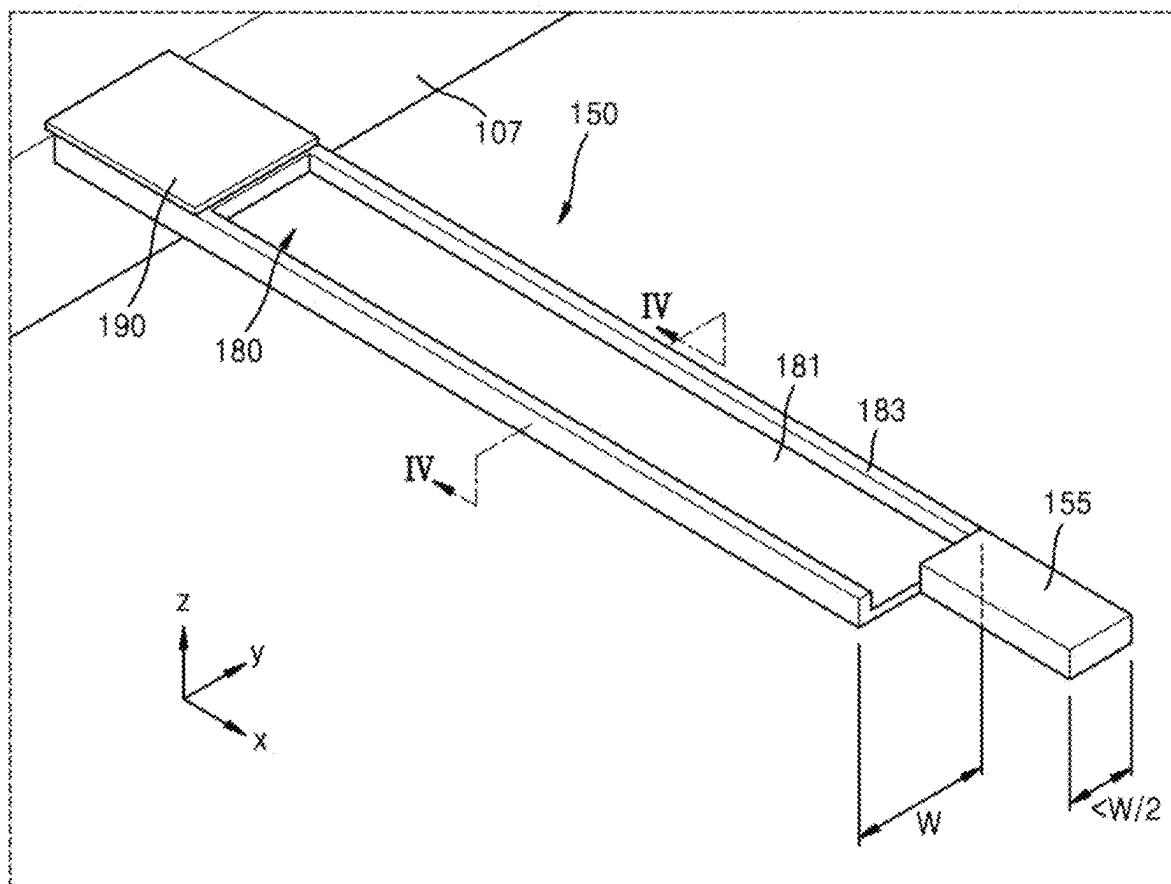


FIG. 3B

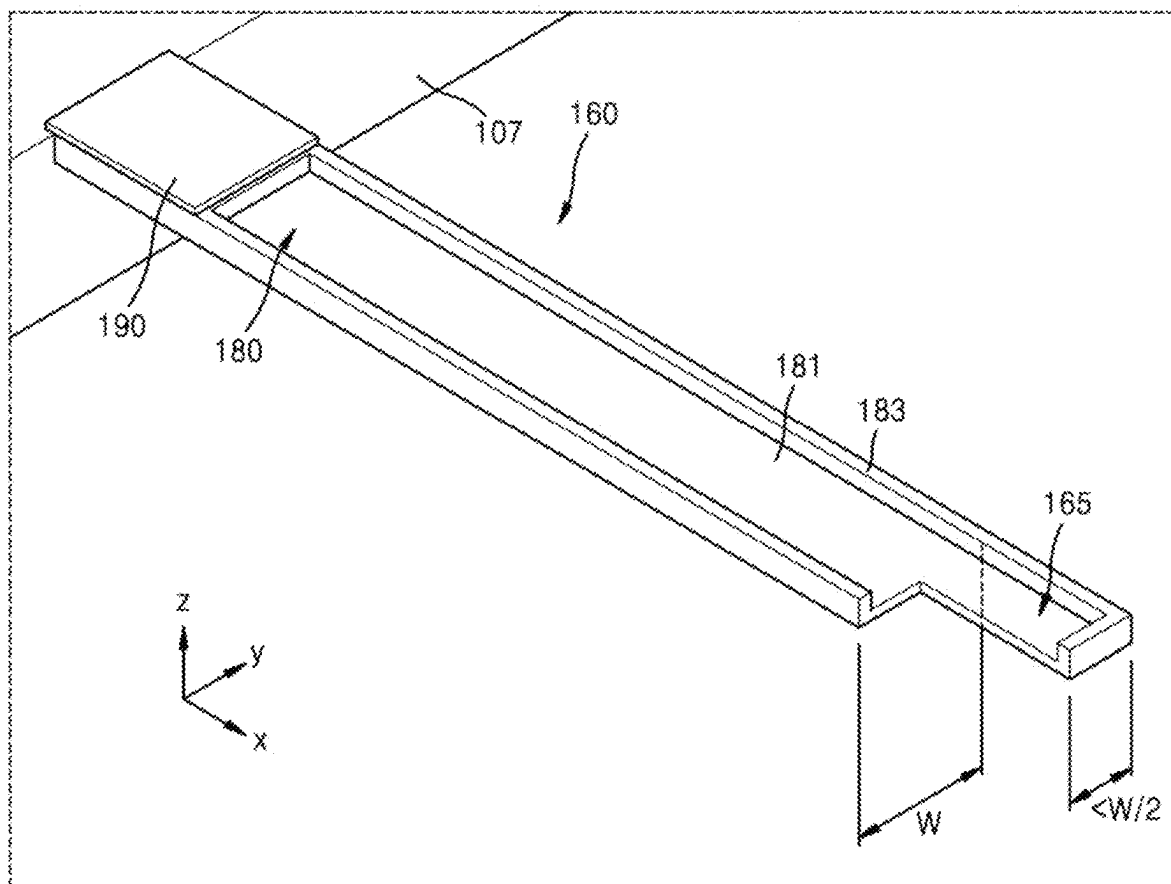


FIG. 3C

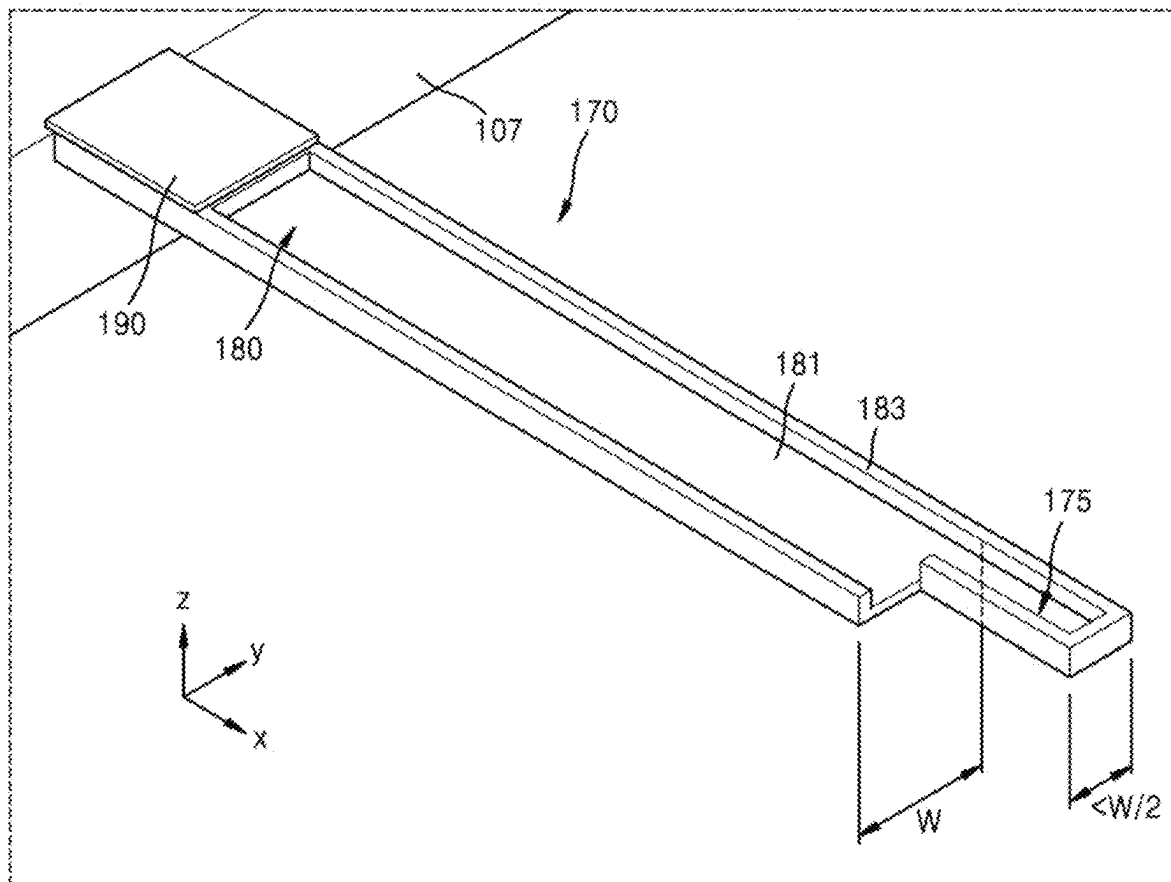


FIG. 3D

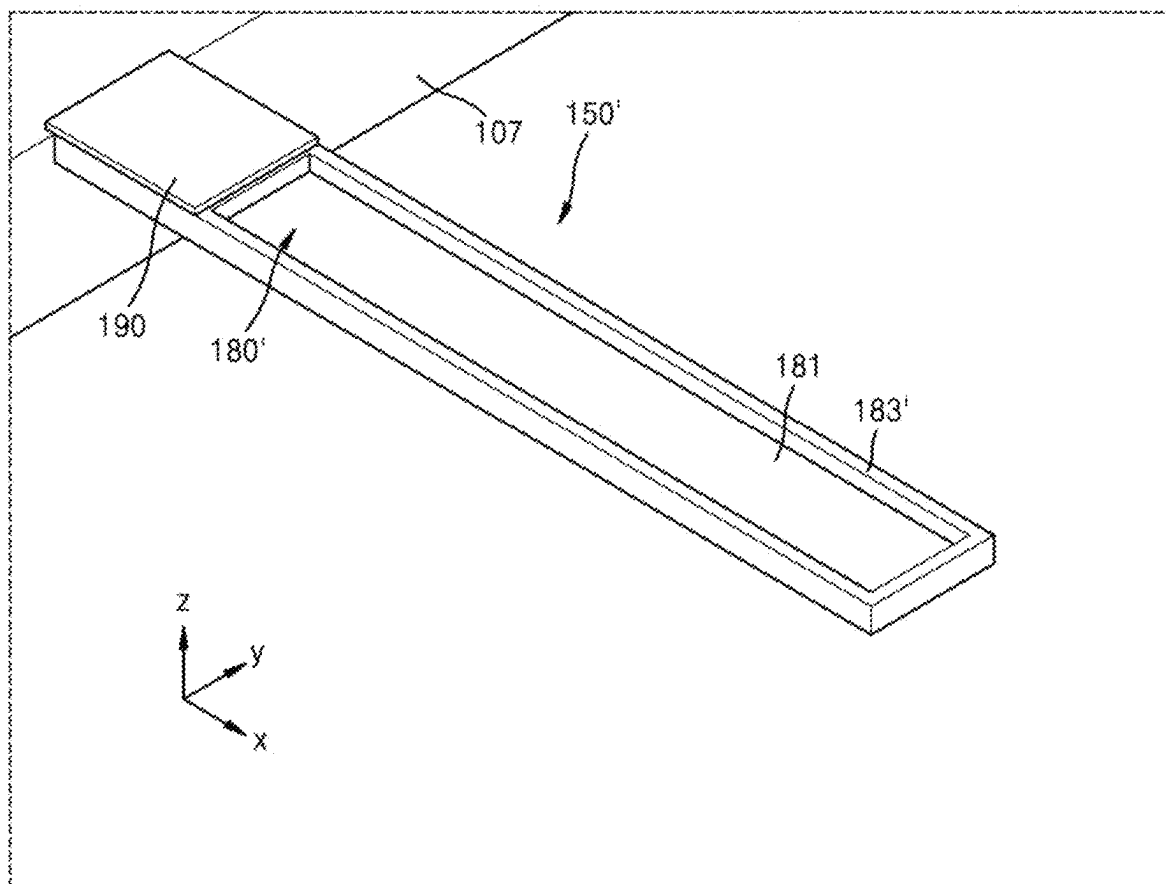


FIG. 4

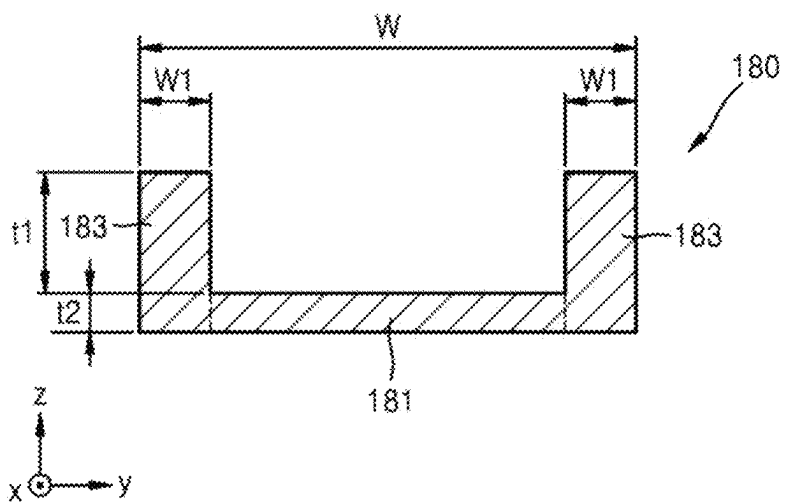


FIG. 5A

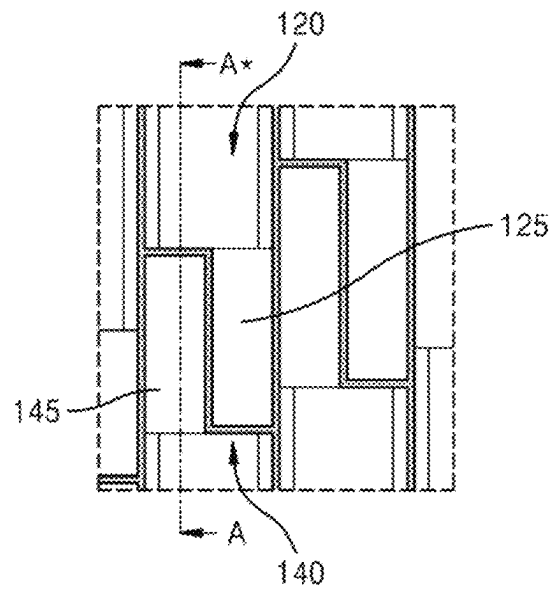


FIG. 5B

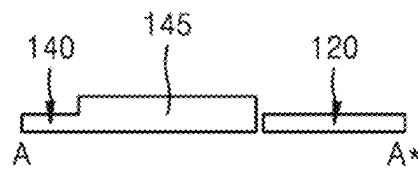


FIG. 6A

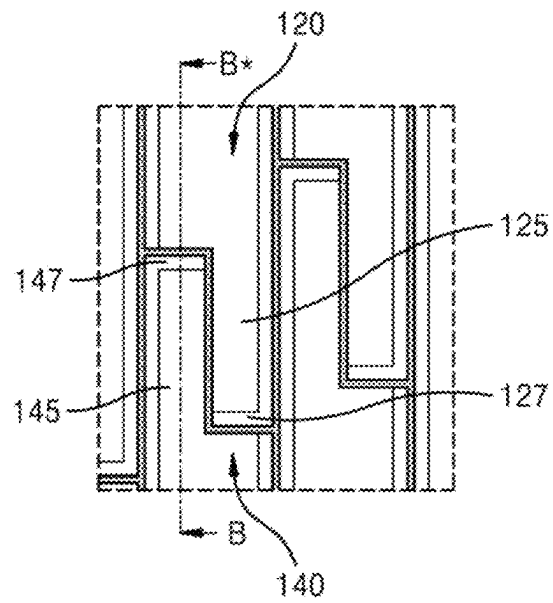


FIG. 6B

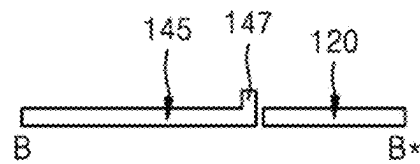


FIG. 7A

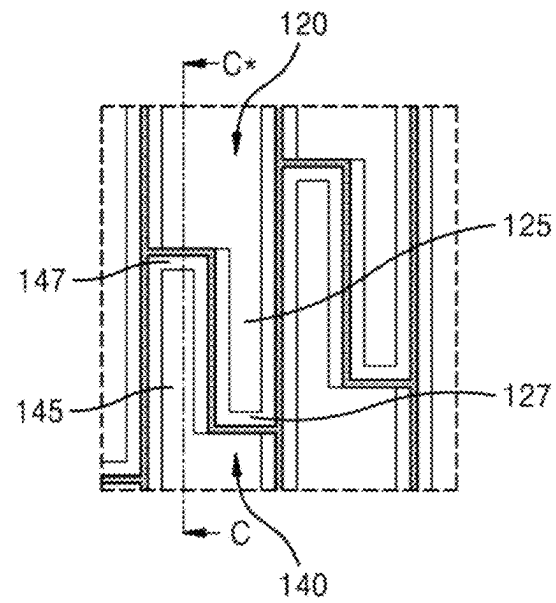


FIG. 7B

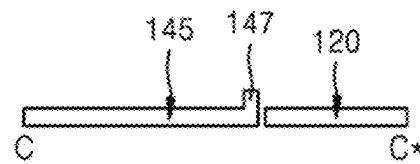


FIG. 8

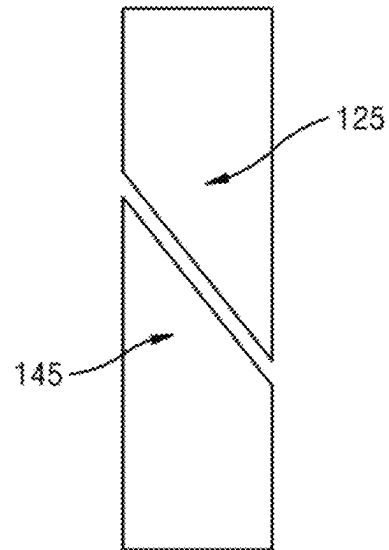


FIG. 9

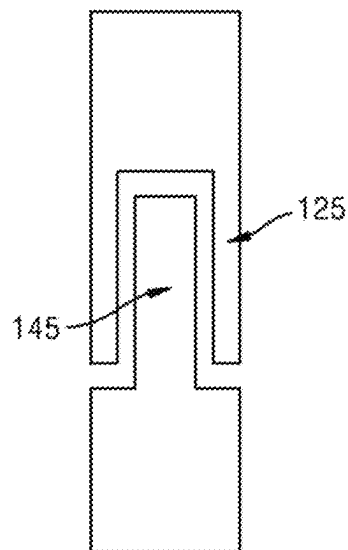


FIG. 10

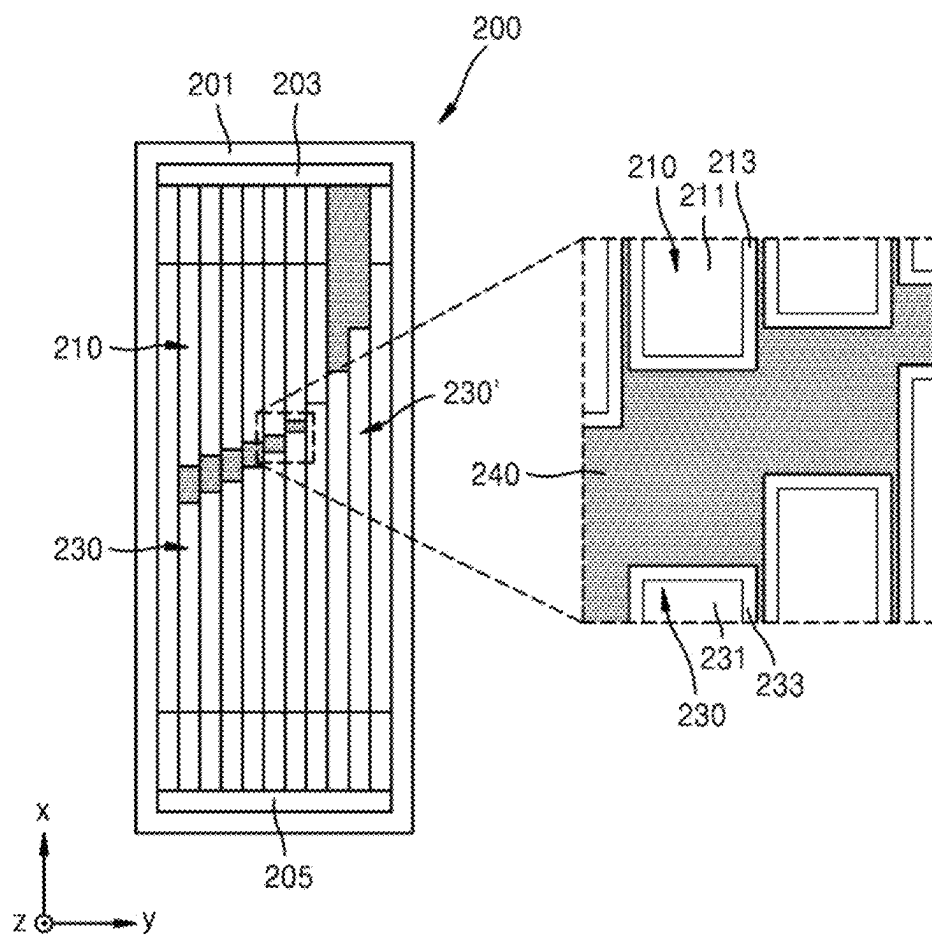


FIG. 11

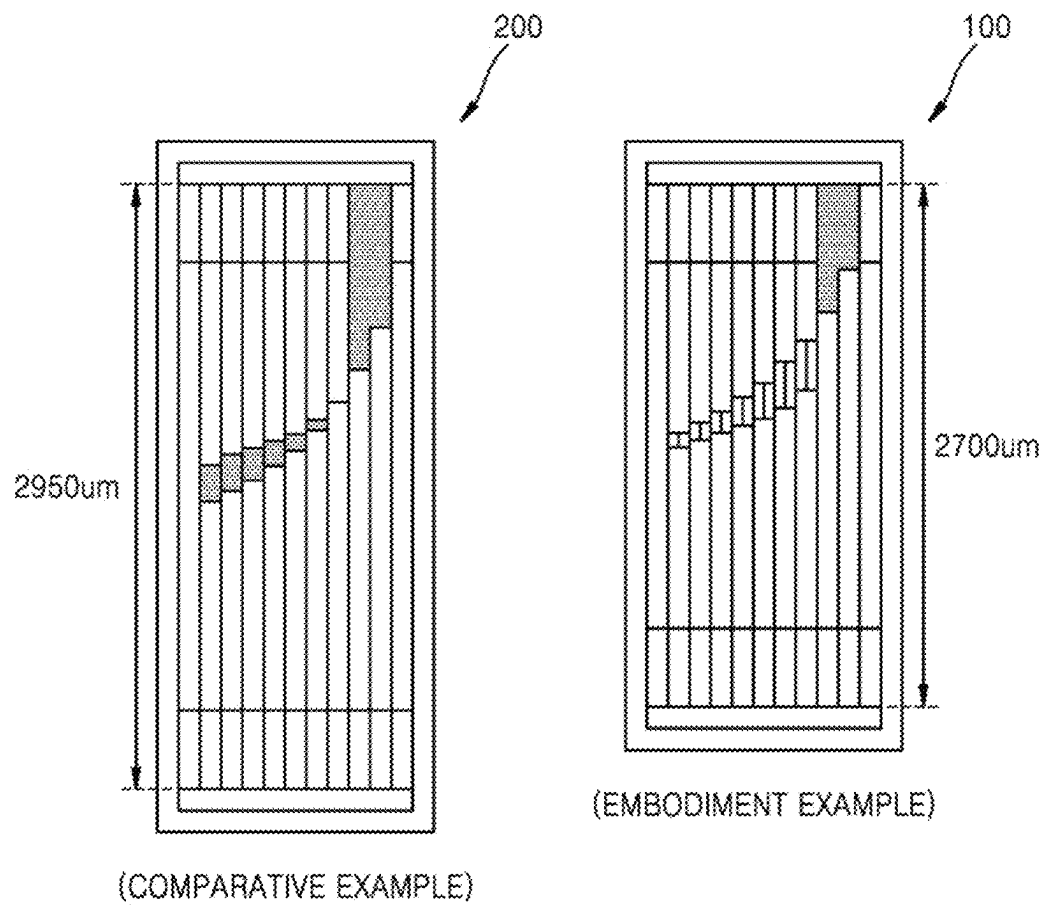


FIG. 12

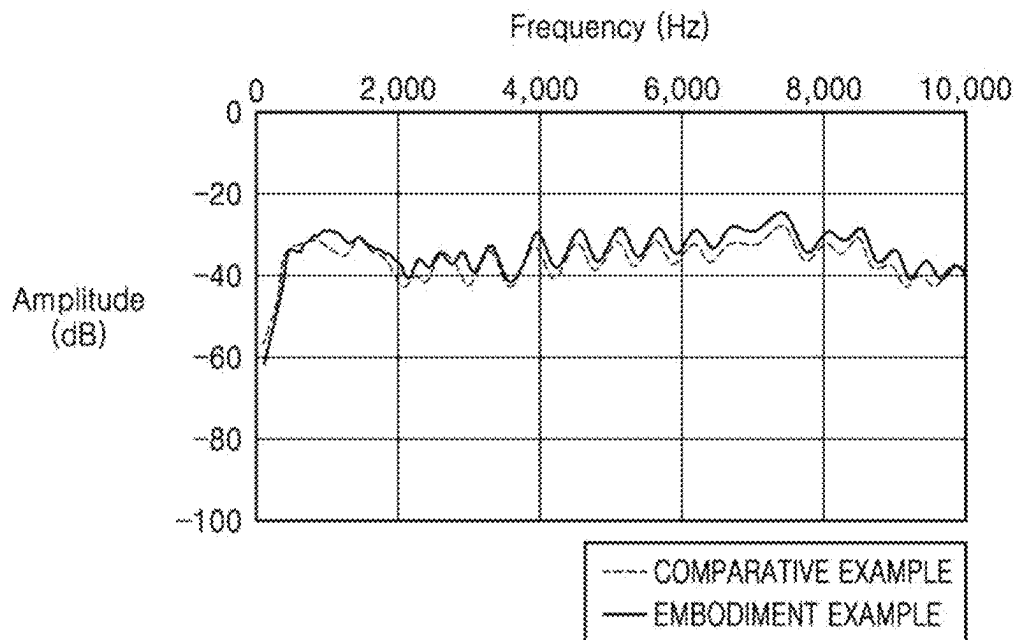


FIG. 13

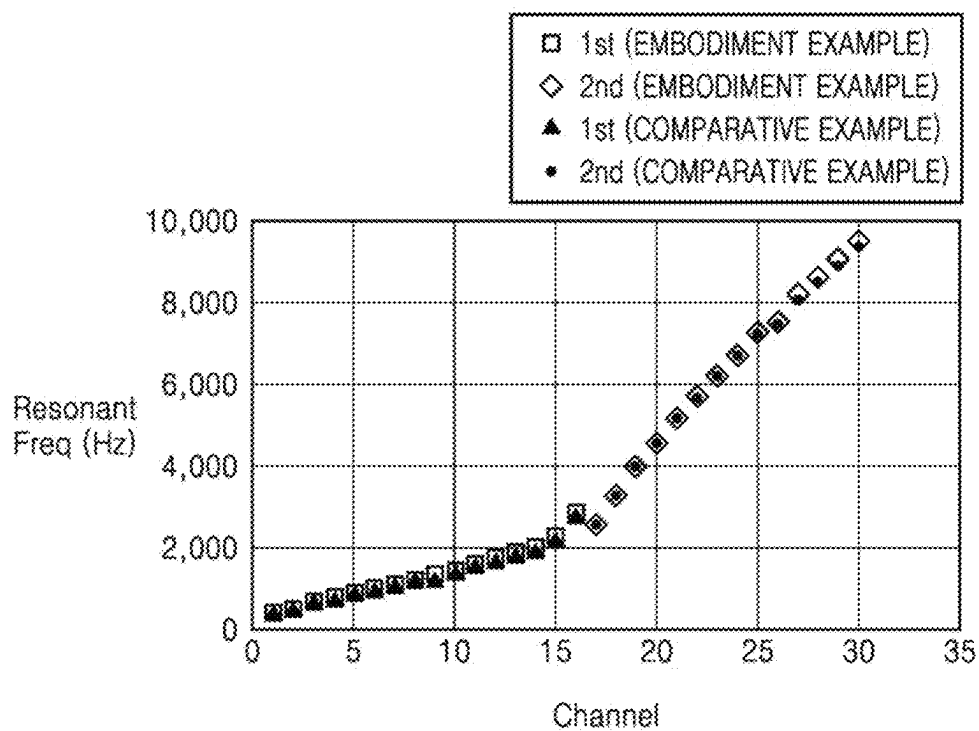


FIG. 14

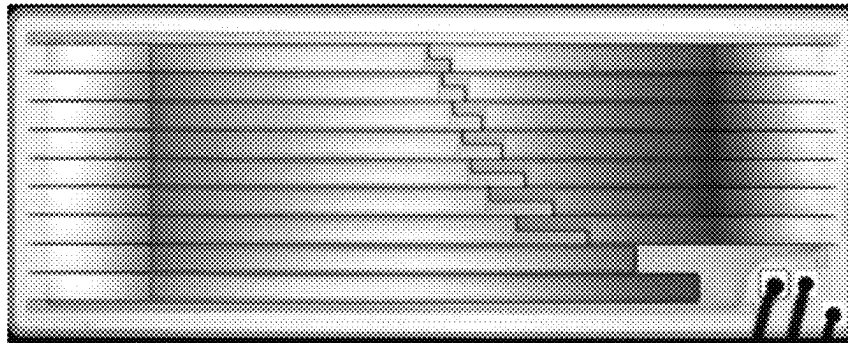


FIG. 15

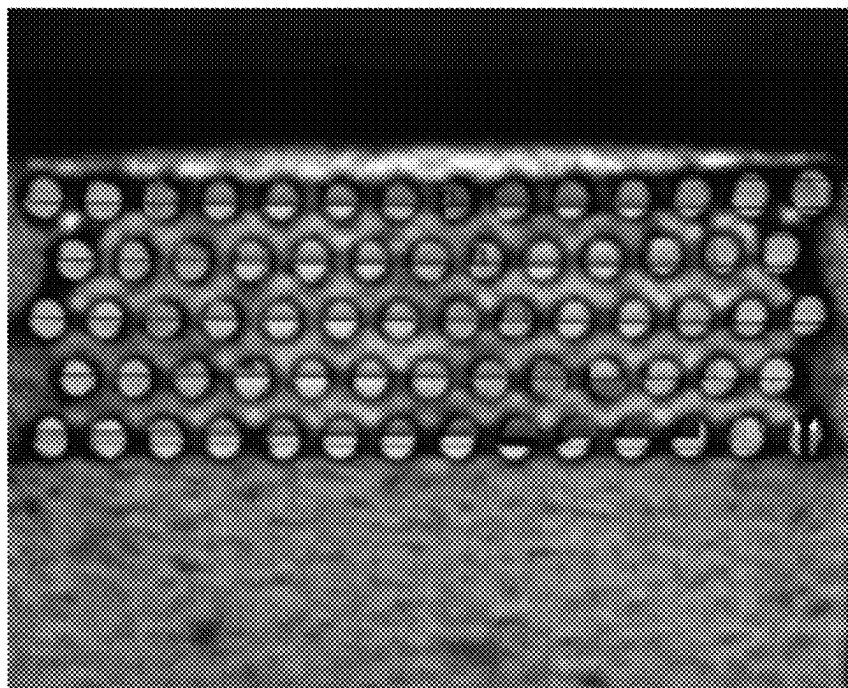


FIG. 16

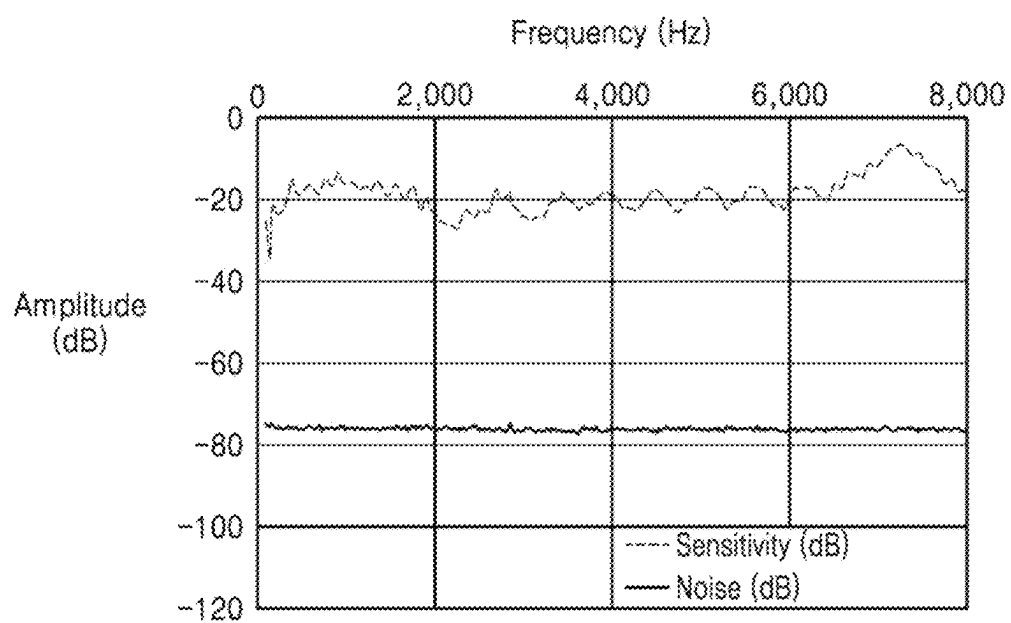


FIG. 17

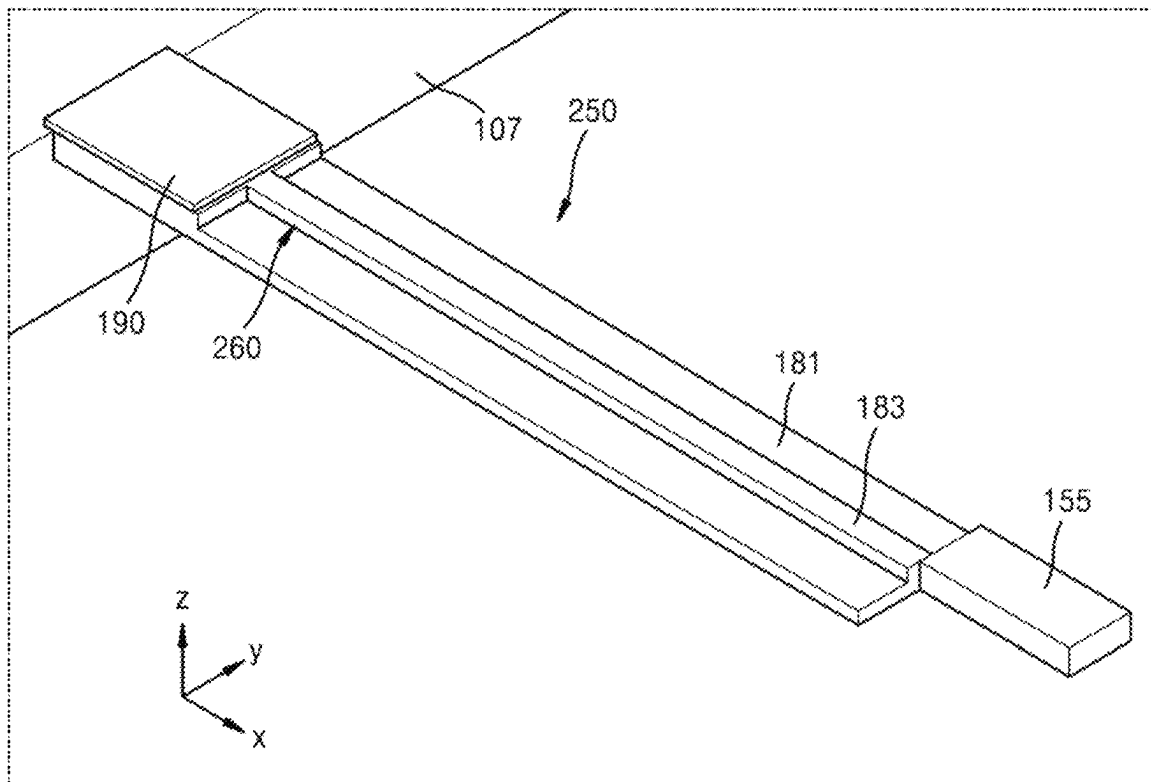


FIG. 18

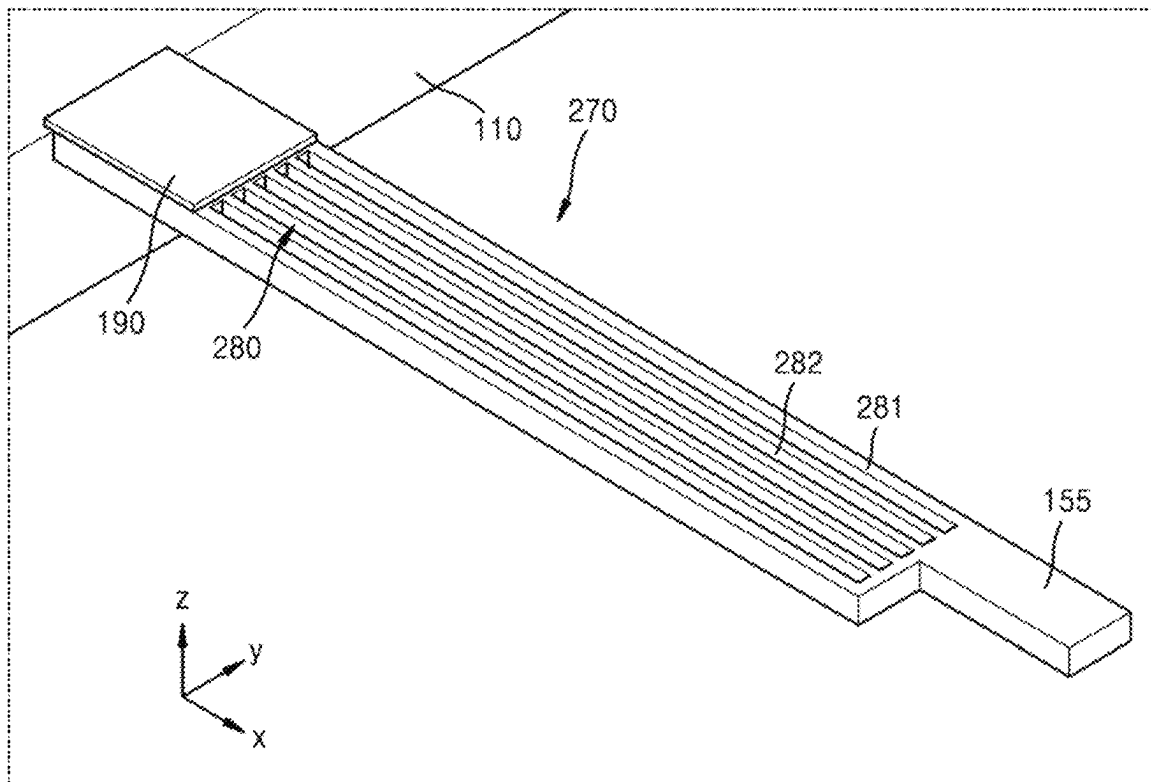


FIG. 19

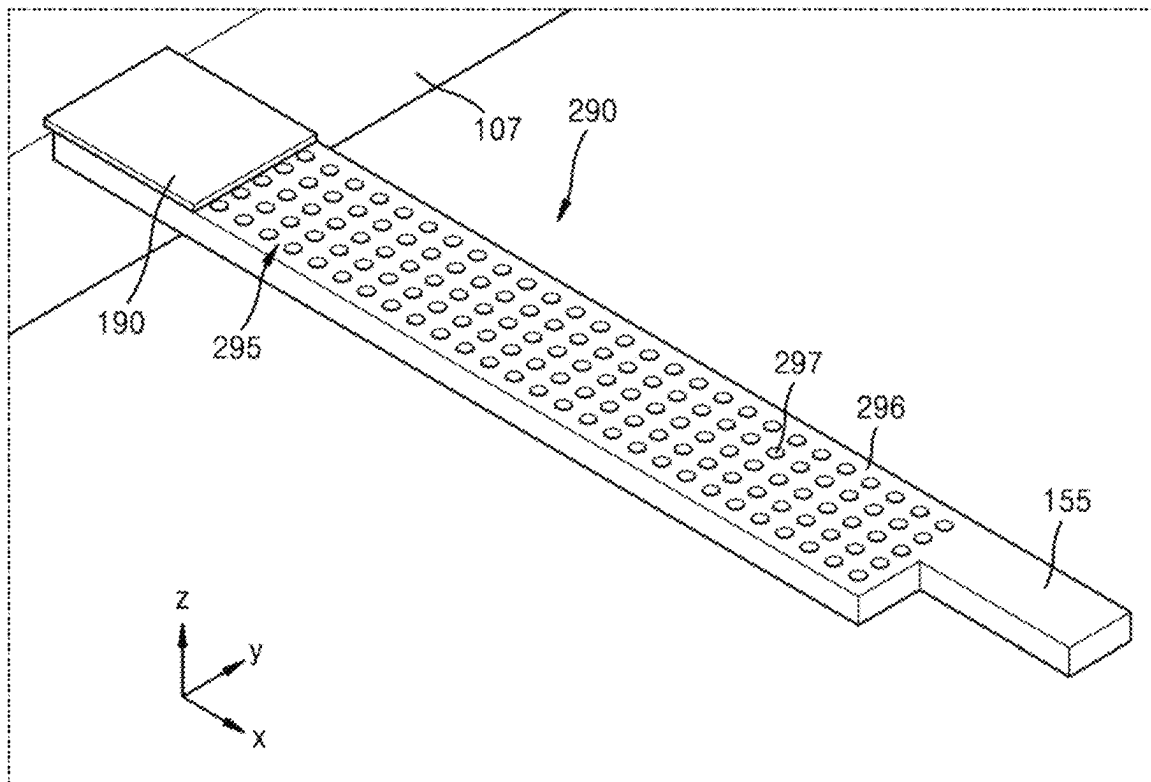


FIG. 20

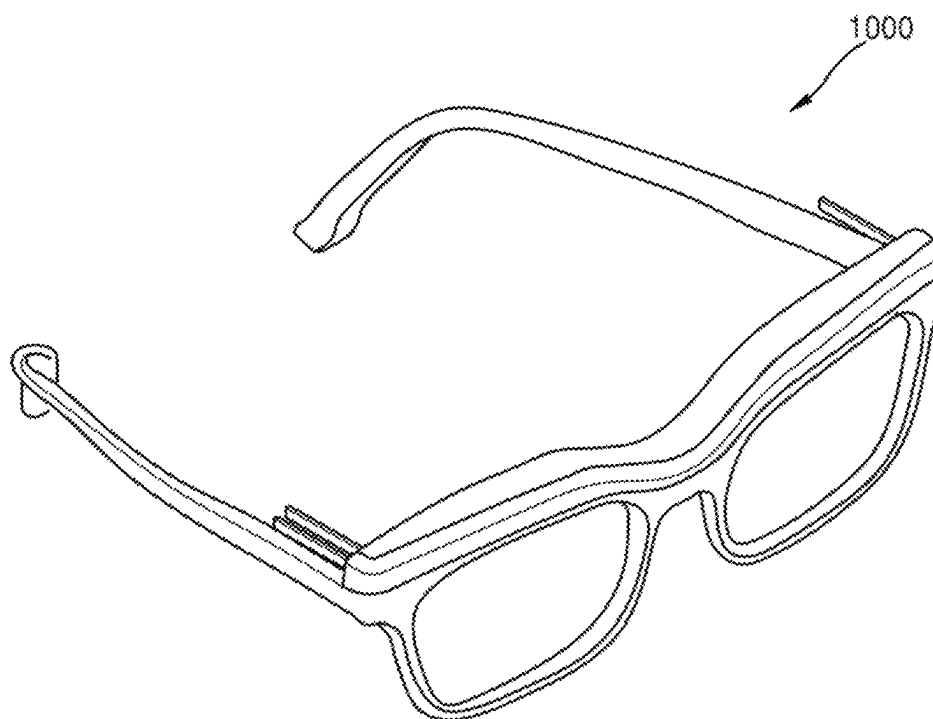


FIG. 21

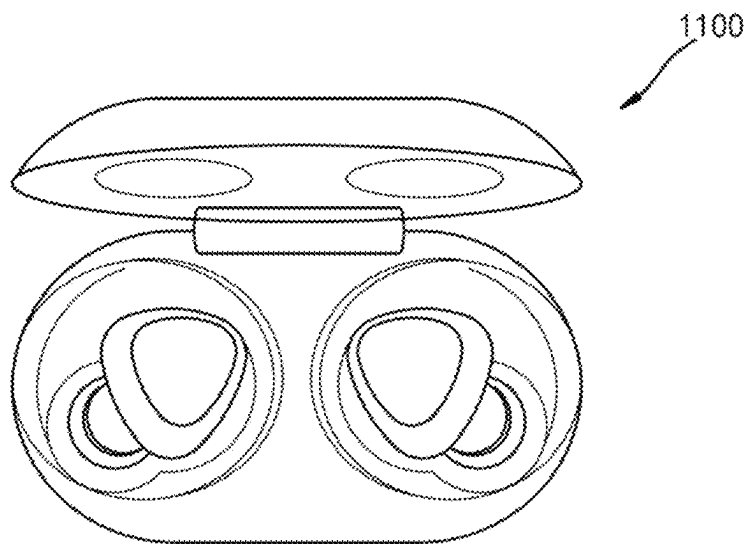


FIG. 22

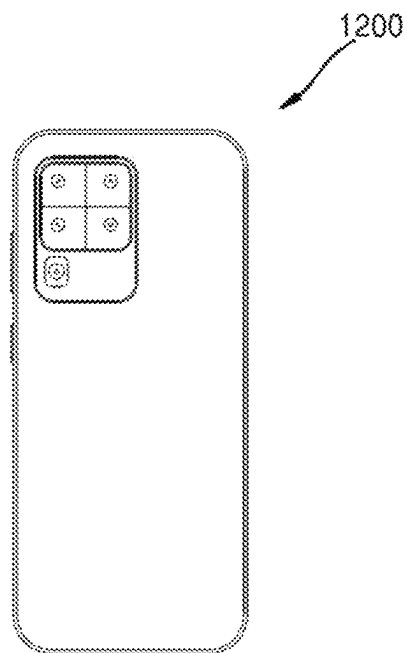


FIG. 23

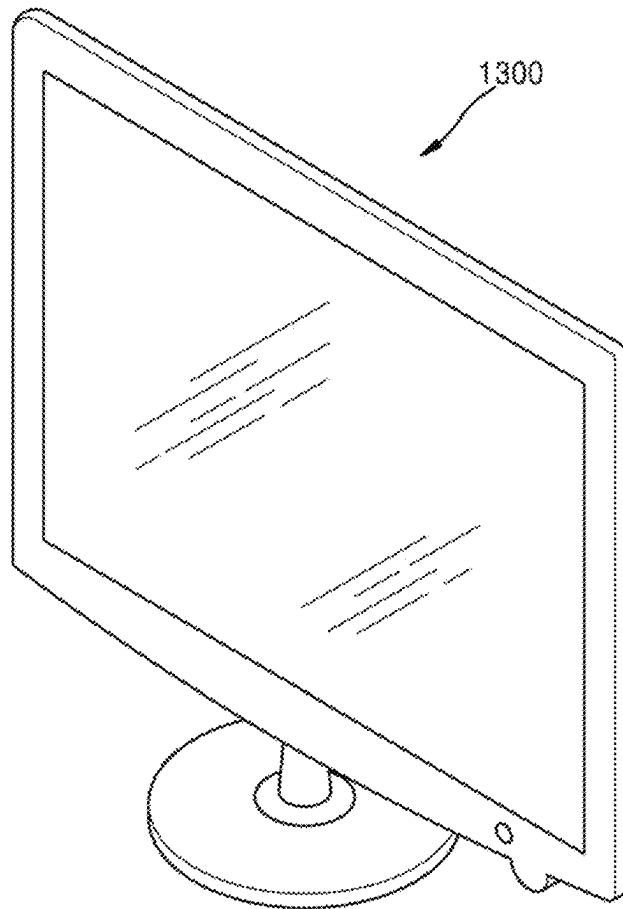


FIG. 24

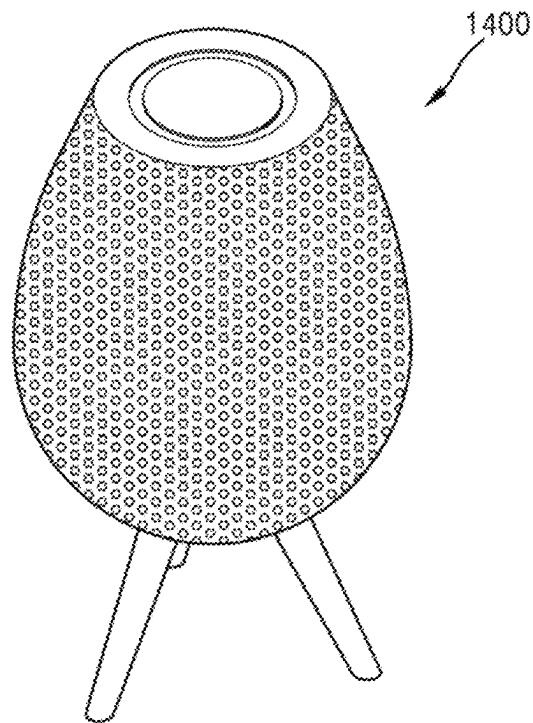
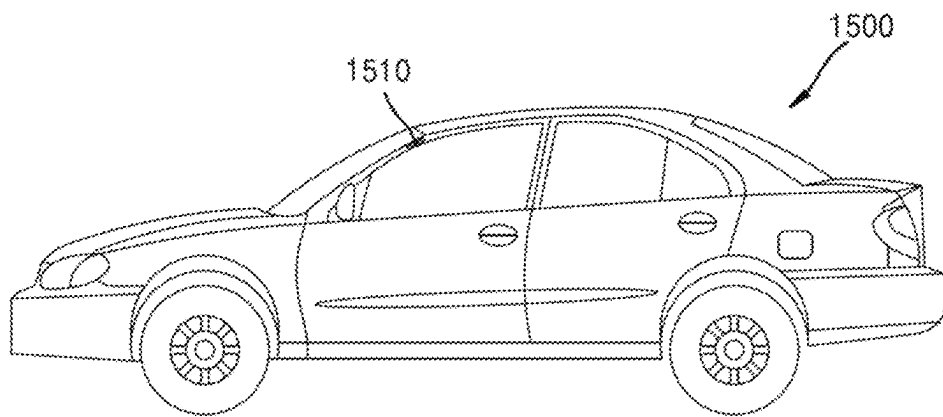


FIG. 25



1

DIRECTIONAL ACOUSTIC SENSOR**CROSS-REFERENCE TO RELATED APPLICATION**

This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0155987, filed on Nov. 12, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND**1. Field**

The present disclosure relates to directional acoustic sensors, and more particularly, to directional acoustic sensors including a resonator that reacts to a pressure difference in an acoustic sound.

2. Description of the Related Art

Utilization of acoustic sensors that are mounted on household appliances, video display devices, virtual reality devices, augmented reality devices, artificial intelligence speakers, etc., to detect a direction of a sound and recognize voices is increasing. Recently, a directional acoustic sensor for detecting an acoustic signal by converting a mechanical motion caused by a pressure difference into an electrical signal has been developed.

SUMMARY

Provided are directional acoustic sensors which may have improved sensitivity by increasing an acoustic resistance.

Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

According to an aspect of an example embodiment, a directional acoustic sensor includes: a support including a first support portion and a second support portion that are separated from each other and face each other; a plurality of first resonators extending in a length direction thereof from the first support portion of the support; and a plurality of second resonators extending in the length direction thereof from the second support portion of the support and facing the plurality of first resonators, wherein each first resonator of the plurality of first resonators has a first end, wherein each second resonator of the plurality of second resonators has a second end, and wherein, in a first resonator arrangement of a region where the plurality of first resonators and the plurality of second resonators face each other, the first ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.

The first end of each first resonator of the plurality of first resonators may have a structure in which a partial width portion of each first resonator extends in the length direction thereof, the second end of each second resonator of the plurality of second resonators may have a structure in which a partial width portion of each second resonator that does not face the partial width portion of a corresponding first resonator extends in the length direction thereof, and a second resonator arrangement may be formed in a region where the plurality of first resonators and the plurality of second resonators face each other in a structure in which the first

2

ends of the plurality of first resonators respectively intersect with the second ends of the plurality of second resonators.

The first ends of the plurality of first resonators and the second ends of the plurality of second resonators may respectively engage with each other.

The first ends of the plurality of first resonators and the second ends of the plurality of second resonators may respectively engage with each other in a male or female form.

The first ends of the plurality of first resonators and the second ends of the plurality of second resonators may respectively engage with each other in an edge shape.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include a base, and a frame protruding from the base and extending in the length direction thereof.

The frame may be integrally formed with the base.

Each of the base and the frame may include silicon.

The frame may be provided on at least one of both sides and an inside of the base.

Each the first ends of the plurality of first resonators and the second ends of the plurality of second resonators may have a plate shape up to a height of the frame.

The base and the frame may extend to each of the first ends of the plurality of first resonators and each of the second ends of the plurality of second resonators.

The first end of each first resonator of the plurality of first resonators may have a structure in which a width portion of a first side of each first resonator extends in the length direction thereof, the second end of each second resonator of the plurality of second resonators may have a structure in which a width portion of a second side of each second resonator extends in the length direction thereof, and the frame may extend to end portions of the first end and the second end.

The first end of each first resonator of the plurality of first resonators may have a structure in which a width portion of a first side of each first resonator extends in the length direction thereof, the second end of each second resonator of the plurality of second resonators may have a structure in which a width portion of a second side of the second resonator that does not face the width portion of the first side of the first resonator extends in the length direction thereof, and the frame may extend to a circumference of the first end and the second end.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include a base and a groove pattern formed in the base to a predetermined depth.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include a base and a plurality of through holes formed in the base and having a size smaller than a wavelength of an audible frequency band.

The directional acoustic sensor may further include at least one third resonator extending in the length direction thereof from the first support portion or the second support portion of the support and not facing the plurality of first resonators or the plurality of second resonators.

The at least one third resonator may include a base and a frame protruding from the base and extending in the length direction thereof.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include: a driving portion configured to move in response to an input sound signal; and a sensing portion configured to detect movement of the driving portion.

3

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include a cantilever beam having a first end fixed to the first support portion and the second support portion and a second end moving freely.

The plurality of first resonators and the plurality of second resonators may have resonant frequencies different from each other.

According to an aspect of an example embodiment, there is provided an electronic device including a directional acoustic sensor including: a support including a first support portion and a second support portion that are separated from each other and face each other; a plurality of first resonators extending in a length direction thereof from the first support portion of the support; and a plurality of second resonators extending in the length direction thereof from the second support portion of the support and facing the plurality of first resonators, wherein each first resonator of the plurality of first resonators has a first end, wherein each second resonator of the plurality of second resonators has a second end, and wherein, in a first resonator arrangement of a region where the plurality of first resonators and the plurality of second resonators face each other, the first ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.

The directional acoustic sensor may further include at least one third resonator extending in the length direction thereof from the first support portion or the second support portion of the support and not facing the plurality of first resonators or the plurality of second resonators.

The at least one third resonator may include a base and a frame protruding from the base and extending in the length direction thereof.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include: a driving portion configured to move in response to an input sound signal, and a sensing portion configured to detect movement of the driving portion.

Each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators may include a cantilever beam having a first end fixed to the first support portion and the second support portion and a second end moving freely.

The plurality of first resonators and the plurality of second resonators may have resonant frequencies different from each other.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of certain example embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a perspective view exemplarily illustrating a directional acoustic sensor according to an example embodiment;

FIG. 2 is a plan view of the directional acoustic sensor of FIG. 1 according to an example embodiment;

FIGS. 3A, 3B and 3C are diagrams of resonators applicable to first and second resonators of FIGS. 1 and 2 according to example embodiments;

FIG. 3D is a diagram of an example of a resonator applicable to a third resonator of FIGS. 1 and 2 according to an example embodiment;

FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 3A according to an embodiment;

4

FIG. 5A is a plan view of an intersecting portion when ends of two facing resonators have plate-shaped ends according to an example embodiment;

FIG. 5B is a cross-sectional view taken along line A-A* of FIG. 5A according to an example embodiment;

FIG. 6A is a plan view of an intersecting portion when ends of two resonators facing each other have an end with a frame extending to the end according to an example embodiment;

FIG. 6B is a cross-sectional view taken along line B-B* of FIG. 6A according to an example embodiment;

FIG. 7A is a plan view of an intersecting portion when ends of two resonators facing each other have an end with a frame extending to a circumference thereof according to an example embodiment;

FIG. 7B is a cross-sectional view taken along line C-C* of FIG. 7A according to an example embodiment;

FIG. 8 is a diagram of an example of forming an intersecting structure in which ends of two resonators facing each other are engaged with each other in an edge shape according to an example embodiment;

FIG. 9 is a diagram of an example of forming an intersecting structure in which ends of two resonators facing each other are engaged with each other in a male/female form according to an example embodiment;

FIG. 10 is a schematic plan view illustrating a directional acoustic sensor according to a comparative example;

FIG. 11 is a diagram of a comparison of a layout size change of a resonator arrangement before and after the application of the intersecting structure of ends of two resonators facing each other according to an example embodiment;

FIG. 12 is a graph showing changes in the frequency response characteristics of resonators before and after the application of the intersecting structure of ends of two resonators facing each other according to an example embodiment;

FIG. 13 is a graph showing resonant frequency characteristics for each channel of the resonator arrangement before and after the application of the intersecting structure of the ends of two resonators facing each other according to an example embodiment;

FIG. 14 is a sample photograph after fabrication of a directional acoustic sensor according to an embodiment;

FIG. 15 is a photograph after packaging the directional acoustic sensor of FIG. 14 according to an example embodiment;

FIG. 16 is a graph showing a result of measuring the frequency response characteristics of a directional acoustic sensor according to a manufactured example embodiment;

FIGS. 17, 18 and 19 are diagrams of resonators that may be applied to two resonators facing each other according to example embodiments; and

FIGS. 20, 21, 22, 23, 24 and 25 are diagrams of electronic devices to which the directional acoustic sensor according to an embodiment is applied as a voice interface device according to example embodiments.

DETAILED DESCRIPTION

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by

5

referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings. In the drawings, like reference numerals refer to like elements, and the size of each component may be exaggerated for clarity and convenience of explanation. The embodiments of the inventive concept are capable of various modifications and may be embodied in many different forms.

When an element or layer is referred to as being “on” or “above” another element or layer, the element or layer may be directly on another element or layer or intervening elements or layers. The singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. When a part “comprises” or “includes” an element in the specification, unless otherwise defined, it is not excluding other elements but may further include other elements.

The term “above” and similar directional terms may be applied to both singular and plural. With respect to operations that constitute a method, the operations may be performed in any appropriate sequence unless the sequence of operations is clearly described or unless the context clearly indicates otherwise. The operations may not necessarily be performed in the order of sequence.

Also, in the specification, the term “units” or “. . . modules” denote units or modules that process at least one function or operation, and may be realized by hardware, software, or a combination of hardware and software.

The connections of lines and connection members between constituent elements depicted in the drawings are examples of functional connection and/or physical or circuitry connections, and thus, in practical devices, may be expressed as replicable or additional functional connections, physical connections, or circuitry connections.

The use of any and all examples, or exemplary language provided herein, is intended merely to better illuminate the inventive concept and does not pose a limitation on the scope of the inventive concept unless otherwise claimed.

A directional acoustic sensor according to an embodiment uses a principle of a resonator that responds to a pressure difference in an acoustic sound, and includes a combination of resonators having different resonant frequencies. Each of the resonators may be formed of a cantilever beam having one end fixed to a support and the other end moving freely, and the resonant frequency of each resonator may be determined by a length of the resonator or a concentration mass of the resonator.

The relationship between the characteristics of the cantilever-type resonator and the resonance frequency may be expressed as Equation (1).

$$\omega_0 = (1.875^2) \sqrt{EI/mL^3} \quad (1)$$

Here, ω_0 indicates resonance frequency, E indicates effective modulus,

$$\left(I = \frac{1}{12} b t^3 \right)$$

6

indicates moment of inertia, b indicates beam width, t is beam thickness, and m indicates effective mass, and L indicates beam length.

When a length of a resonator is the same and a mass is changed, a bandwidth of the resonator may be reduced. This is because if the mass is increased to lower a frequency, a Q-factor increases and the bandwidth decreases. Therefore, in order to implement the same bandwidth characteristics even though the frequencies are different from each other, the adjustment of a resonant frequency by changing the length of the resonator is suitable for designing a directional acoustic sensor with a wide frequency band. When a required resonance frequency is designed, the number and length of resonators may be determined and the size of a device may be determined. However, when resonators having lengths different from each other are arranged, a through hole may be generated. The through hole may reduce an acoustic resistance, thereby reducing a pressure difference in acoustic sound, and may be a factor of lowering sensitivity.

According to the directional acoustic sensor according to an embodiment, a through-hole portion of a cavity may be minimized by forming a resonator arrangement in which a plurality of resonators face each other, and forming the resonator arrangement in a structure in which ends of two facing resonators intersect each other. By reducing or almost eliminating a through-hole portion by such an arrangement of the resonator, a decrease in sensitivity due to a decrease in acoustic resistance may be prevented, and the element size of the directional acoustic sensor may also be reduced.

FIG. 1 is a perspective view illustrating a directional acoustic sensor according to an example embodiment. FIG. 2 is a plan view of the directional acoustic sensor of FIG. 1 according to an example embodiment.

Referring to FIGS. 1 and 2, the directional acoustic sensor 100 includes a support 101 and a plurality of first resonators 110 and a plurality of second resonators 130 provided on the support 101. The directional acoustic sensor 100 according to the embodiment may include at least one third resonator 130' provided on the support 101, although in some embodiments, the at least one third resonator 130' is not included.

The support 101 is formed to include a cavity that penetrates therethrough, and includes first and second support portions 103 and 105 facing each other with the cavity therebetween. A separation distance between the first and second support portions 103 and 105 may correspond to a width of the cavity in a length direction of the directional acoustic sensor 100.

The support 101 may be, for example, a silicon substrate, but is not limited thereto. The plurality of first resonators 110 and the plurality of second resonators 130 may be arranged in the cavity of the support 101 in a predetermined shape. For example, the plurality of first resonators 110 and the plurality of second resonators 130 may be arranged to form a resonator arrangement to face each other in the cavity of the support 101. The at least one third resonator 130' may be arranged in a predetermined shape in the cavity of the support 101 so as not to face the first resonator 110 or the second resonator 130.

One end of each of the plurality of first resonators 110 may be fixed to the first support portion 103 of the support 101 to form a planarly parallel arrangement in the cavity of the support 101. One end of each the plurality of second resonators 130 may be fixed to the second support portion 105 of the support 101 to form the planarly parallel arrangement in the cavity of the support 101. One end of the at least one third resonator 130' may be fixed to, for example, the second support portion 105 of the support 101 to form the

planarly parallel arrangement in the cavity of the support **101**. Here, one end of the third resonator **130'** may be fixed to the first support portion **103** of the support **101**.

Each of the plurality of first resonators **110** may extend in the length direction thereof from the first support portion **103** of the support **101**. Each of the plurality of second resonators **130** may extend in the length direction thereof with respect to the second support portion **105** of the support **101**, and may be arranged to face the plurality of first resonators **110**. The at least one third resonator **130'** may extend in the length direction thereof from the second support portion **105** towards the first support portion **103**.

FIGS. **1** and **2** show an example arrangement in which the plurality of first resonators **110** and the plurality of second resonators **130** face each other, and the third resonator **130'** that does not face the first resonator **110** is included.

As another example, the plurality of first resonators **110** and the plurality of second resonators **130** are arranged to face each other, and the at least one third resonator **130'** may be arranged not to face the second resonator **130**. As another example, the number of first resonators **110** is the same as the number of second resonators **130**, the plurality of first resonators **110** and the plurality of second resonators **130** are all arranged to face each other, and the at least one third resonator **130'** that does not face the first resonator **110** or the second resonator **130** is not included. Hereinafter, a case in which the directional acoustic sensor **100** according to an embodiment further includes the at least one third resonator **130'** that does not face the first resonator **110**, and the third resonator **130'** is arranged parallel to the second resonator **130** will be described as an example.

Here, the length direction is a direction corresponding to the length of each of the resonators, that is, the first resonator **110**, the second resonator **130**, and the third resonator **130'**, may correspond to a direction from the first support portion **103** to the second support portion **105** or correspond to a direction from the second support portion **105** to the first support portion **103**. That is, the length direction may correspond to a direction (x-axis direction) crossing the first support portion **103** and the second support portion **105**.

Each of the plurality of first resonators **110** may constitute a cantilever beam in which one end is fixed to the first support portion **103** of the support **101** and the other end freely moves. Each of the plurality of second resonators **130** and the at least one third resonator **130'** may form a cantilever beam in which one end is fixed to the second support portion **105** of the support **101** and the other end freely moves. FIGS. **1** and **2** illustrate a case in which the plurality of first resonators **110**, the plurality of second resonators **130**, and the at least one third resonator **130'** move in a z-axis direction. FIG. **1** exemplarily illustrates a state in which some portion of the second resonator **130** are moved by an input acoustic sound.

Each of the plurality of first resonators **110** includes a driving portion **120** configured to vibrate in response to an input sound signal, and a sensing portion configured to detect a movement of the driving portion **120**. Each of the plurality of second resonators **130** includes a driving portion **140** configured to vibrate in response to an input sound signal, and a sensing portion configured to detect a movement of the driving portion **140**. The driving portions **120** and **140** of each of the plurality of first resonators **110** and the plurality of second resonators **130** may be configured as a cantilever beam.

One end of the driving portion **120** of each of the plurality of first resonators **110** may be fixed to the first support portion **103** of the support **101**, and extend in the length

direction of the first resonator **110** from the first support portion **103** toward the cavity. The driving portion **120** may be integrally formed with the support **101**, and in this case, the driving portion **120** may include, for example, silicon. However, the present embodiment is not limited thereto, and the driving portion **120** may not be integrally formed with the support **101**. The driving portion **120** of each of the plurality of first resonators **110** may have a first end **125** at an end thereof.

One end of a driving portion **140** of each of the plurality of second resonators **130** and the at least one third resonator **130'** may be fixed to the second support portion **105** of the support **101**, and may extend in the length direction (an x-axis direction in FIGS. **1** and **2**) of the second and third resonators **130** and **130'** from the second support portion **105** toward the cavity. The driving portion **140** may be integrally formed with the support **101**, and in this case, the driving portion **140** may include, for example, silicon. However, the present embodiment is not limited thereto, and the driving portion **140** may not be integrally formed with the support **101**. The driving portion **140** of each of the plurality of second resonators **130** may have a second end **145** at the end thereof.

An effective length of the first resonator **110** is a vibrating length of the driving portion **120**, and may include a length of the first end **125**. An effective length of the second resonator **130** is a vibrating length of the driving portion **140**, and may include a length of the second end **145**.

The directional acoustic sensor **100** according to the embodiment is configured such that the first end **125** and the second end **145** of the first resonator **110** and the second resonator **130** that face each other form an intersecting structure. The length of each of the first end **125** and the second end **145** for the intersecting structure may be formed up to approximately half of the effective length of each of the first resonator **110** and the second resonator **130**. That is, the length of each of the first and second ends **125** and **145** forming the intersecting structure of the first and second resonators **110** and **130** facing each other may be up to about half the length of each of the first and second resonators **110** and **130**.

Sensing portions **111** and **131** may be provided on one surface of each of the driving portions **120** and **140** to detect the movement of the driving portions **120** and **140**. The sensing portions **111** and **131** may include, for example, a piezoelectric element that generates electric energy by deformation of the piezoelectric material. In this case, each of the sensing portions **111** and **131** may include two electrodes and a piezoelectric layer provided between the two electrodes.

The first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** disposed to face each other may respectively correspond to the end of the cantilever beam, which is the driving portion **120** or **140**.

The directional acoustic sensor **100** according to the embodiment includes a resonator arrangement region in which the first resonator **110** and the second resonator **130** face each other, and in the resonator arrangement to face each other, the first resonator **110** and the second resonator **130** may be arranged to form a structure in which the first end **125** and the second end **145** intersect each other.

For example, the first resonator **110** and the second resonator **130** that face each other may have a resonator arrangement such that the first end **125** of the first resonator **110** has a structure in which a width portion of one side of the first resonator **110** extends in the length direction thereof, the second end **145** of the second resonator **130** has a

structure in which a width portion of the other side of the second resonator **130** that does not face the width portion of the one side of the first resonator **110** extends in the length direction thereof, and the first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** have an intersecting structure.

In this case, the first end **125** and the second end **145** may be engaged with each other. The first resonator **110** and the second resonator **130** may be arranged facing each other such that the first end **125** and the second end **145** are engaged with each other, but direct collision does not occur and a gap between them is minimized. In this way, when the first end **125** and the second end **145** of the first resonator **110** and the second resonator **130** that face each other form an intersecting structure engaged with each other, in the arrangement region where the first resonator **110** and the second resonator **130** face each other, a through-hole portion of the cavity may be reduced or almost absent, and thus, a decrease in sensitivity due to a decrease in an acoustic resistance may be prevented, the sensitivity for sensing acoustic sound may be improved, and the device size of the directional acoustic sensor **100** may be reduced.

The first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** disposed to face each other may have the same length or different lengths, and a sum of widths of the first end **125** and the second end **145** may be less than half of a sum of a maximum width of the first resonator **110** and a maximum width of the second resonator **130**.

For example, consider a case in which the maximum width of the first resonator **110** and the second resonator **130** disposed to face each other is equal to W . In this case, when the first resonator **110** and the second resonator **130** facing each other are arranged to form an intersecting structure in which the first end **125** and the second end **145** are engaged with each other, the sum of the width of the first end **125** and the width of the second end **145** may be less than or equal to W such that a gap between the first end **125** and the second end **145** is minimized without collision between the first end **125** and the second end **145**. In this case, the first end **125** and the second end **145** of the first resonator **110** and the second resonator **130** facing each other may have the same length or different lengths, and the width of the first end **125** may be the same as or different from the width of the second end **145**. For example, the first end **125** and the second end **145** of the first resonator **110** and the second resonator **130** facing each other may be the same width, and may have a width equal to or less than $W/2$, respectively. When a gap between the first end **125** and the second end **145** intersecting each other is large, an acoustic resistance may be reduced, and thus, the shape, length, and width of the first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** arranged to intersect each other may be determined such that the decrease in sensitivity is prevented by suppressing the decrease in acoustic resistance and a desired resonant frequency is realized without collision between the first end **125** and the second end **145** engaged with each other.

FIGS. 1 and 2 exemplarily illustrate an arrangement in which the plurality of first resonators **110** and the plurality of second resonators **130** are disposed to face each other, and the at least one third resonator **130'** does not face the first resonator **110**. In the case of the third resonator **130'** that does not face the first resonator **110**, the second end **145** having a different shape may not be provided as exemplarily shown in FIGS. 1 and 2. For example, the third resonator **130'** may be formed to have a constant width. As another

example, similar to the second resonator **130** facing the first resonator **110**, even in the case of the third resonator **130'** not facing the first resonator **110**, the second end **145** having a different shape may be provided.

The plurality of first resonators **110**, the plurality of second resonators **130**, and the at least one third resonator **130'** may sense sound frequencies of different bands. That is, the plurality of first resonators **110** may be configured to have different resonance frequencies. To this end, the plurality of first resonators **110** may have dimensions different from each other. For example, the plurality of first resonators **110** may have lengths, widths, or thicknesses different from each other. Also, the plurality of second resonators **130** may be configured to have different resonance frequencies. To this end, the plurality of second resonators **130** may have dimensions different from each other. For example, the plurality of second resonators **130** may have lengths, widths, or thicknesses different from each other. In addition, the shapes, lengths, and widths of the first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** arranged to intersect each other may be determined so as to realize a desired resonant frequency without collision between the first end **125** and the second end **145** that intersect each other. Also, the at least one third resonator **130'** may be configured to have a different resonance frequency. To this end, the at least one third resonator **130'** may have a different dimension. For example, the at least one third resonator **130'** may have a different length, width, or thickness.

FIGS. 1 and 2 illustrate, as an example, a case in which the plurality of first resonators **110** have different lengths from each other, the plurality of second resonators **130** have different lengths from each other, the maximum widths of the first resonator **110** and the second resonator **130** are equal to W , and the first end **125** and the second end **145** have the same width of $W/2$ or less. The directional acoustic sensor **100** according to the present embodiment may have a directivity, for example, bi-directionality in a z -axis direction.

FIGS. 3A to 3C illustrate various examples of resonators **150**, **160**, **170** that may be applied as the first resonator **110** and the second resonator **130** facing each other in FIGS. 1 and 2 according to embodiments. FIG. 3D illustrates an example of a resonator **150'** that may be applied as the third resonator **130'** that does not face the first resonator **110** in FIGS. 1 and 2 according to an embodiment. FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 3A, which corresponds to a cross-sectional view of the resonators **150**, **150'**, **160**, and **170** of FIGS. 3A to 3D according to an embodiment. As a third resonator **130'** that does not face the first resonator **110**, the resonators **150**, **160**, and **170** of FIGS. 3A to 3C may be applied instead of the resonator **150'** of FIG. 3D.

Referring to FIGS. 3A to 3D, each of the resonators **150**, **150'**, **160**, and **170** includes a driving portion **180** that vibrates in response to an acoustic signal, and a sensing portion **190** detecting a movement of the driving portion **180**.

The driving portion **180** of each of the resonators **150**, **150'**, **160**, and **170** is configured of a cantilever beam, and may extend in a length direction (x -axis direction in FIGS. 1 and 2) of the resonators **150**, **150'**, **160**, **170** from a support **107** (corresponding to the first support portion **103** or the second support portion **105** in FIGS. 1 and 2) toward a cavity. One end of the driving portion **180** may be fixed to the support portion **107** of a support. The driving portion **180** may be integrally formed with the support, and in this case,

11

the driving portion **180** may include, for example, silicon. However, the present embodiment is not limited thereto, and the driving portion **180** may not be formed integrally with the support. The sensing portion **190** may be provided on one surface of the driving portion **180** to detect the movement of the driving portion **180**. The sensing portion **190** may include a piezoelectric element that generates electric energy by deformation of the piezoelectric material. In this case, the sensing portion **190** may include two electrodes and a piezoelectric layer provided between the two electrodes.

The driving portion **180** of the resonators **150**, **160**, **170** may include a base **181** and a frame **183** protruding from the base **181** and extending in the length direction thereof. The driving portion **180'** of the resonators **150'** may include a base **181** and a frame **183'** protruding from the base **181** and extending in the length direction thereof.

Referring to FIGS. 3A to 3C, the driving portions **180** of the resonators **150**, **160**, and **170** respectively have ends **155**, **165**, and **175** at end portions thereof (corresponding to the first end **125** and the second end **145** in FIGS. 1 and 2). Referring to FIG. 3D, the driving portion **180'** of the resonator **150'** may be formed such that the frame **183'** extends also to an end portion of the base **181**. In the driving portion **180/180'** of the resonators **150**, **160**, and **170/150'**, the frame **183/183'** may be integrally formed with the base **181**, and the ends **155**, **165**, and **175** may also be integrally formed with the base **181** and the frame **183**. The base **181** and the frame **183/183'** may include silicon, and the ends **155**, **165**, and **175** may also include silicon.

The frame **183/183'** may be provided on at least one of both sides and an inner side of the base **181**. The frame **183/183'**, for example, as shown in FIGS. 3A to 3D, may be provided on both sides of the base **181**. Here, the frame **183/183'** may be continuously provided in parallel to the length direction (x-axis direction in FIGS. 1 and 2) of the resonators **150**, **160**, and **170/150'**. For example, the frame **183/183'** may be continuously provided along both edges of the base **181**.

FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 3A as an example. Referring to FIG. 4, the base **181** may have a predetermined width W, and the frame **183** may have a predetermined width W1. In addition, the frame **183** may have a predetermined height t1, and the base **181** may have a predetermined thickness t2. The predetermined thickness t2 of the base **181** may be less than the height t1 of the frame **183**, but the present embodiment is not limited thereto.

By configuring the driving portion **180** of the resonator to include the base **181** and the frame **183** extending in the length direction thereof to protrude from the base **181**, compared with a plate-type resonator having a thickness corresponding to a sum of the thickness of the base **181** and the height of the frame **183**, a mass thereof may be reduced while maintaining the same resonant frequency of the resonator. Accordingly, a bandwidth may be widened by lowering the quality factor of the resonator, and a flatness characteristic and sound quality of the directional acoustic sensor **100** including the resonator arrangement may be improved.

Here, in order to implement a wide frequency range by using a limited number of resonators and to improve flatness characteristics, each of the resonators is required to have a wide bandwidth, and for this purpose, a quality factor of the resonators must be small. The quality factor Q of the resonator may be expressed by Equation (2) below.

$$Q=(mxf)/c$$

(2)

12

Here, m is an effective mass, f is a resonance frequency, and c is a damping coefficient.

Referring to Equation (2), in order to reduce the quality factor while maintaining the resonant frequency of the resonator constant, it is required to reduce the mass of the resonator. When the mass of the resonator is reduced by reducing the thickness of the resonator, the length of the resonator must also be reduced to maintain the resonant frequency, and accordingly, an area of a portion receiving pressure from the input sound signal is reduced, thereby reducing the sensitivity of the directional acoustic sensor **100**.

However, as in the directional acoustic sensor **100** according to the embodiment, by applying the resonators **150**, **160**, and **170/150'** having the driving portion **180/180'** including the base **181** and the frame **183/183'** extending in the length direction thereof to protrude on the base **181** to the plurality of first resonators **110**, the plurality of second resonators **130**, and the at least one third resonator **130'**, the mass of each resonator may be reduced while maintaining a resonance frequency constant, and accordingly, compared to a conventional plate-type resonator, the quality factor may be reduced and the bandwidth may be increased.

As described above, in the directional acoustic sensor **100** according to the embodiment, by configuring each of the plurality of first resonators **110**, the plurality of second resonators **130**, and the at least one third resonator **130'** to include a frame extending in the length direction thereof to protrude from the base, each resonator may have a wide bandwidth while maintaining a resonant frequency, thereby improving sensitivity and flatness characteristics, and as a result, the number of resonators may be reduced, thereby improving price competitiveness. In addition, sound quality may be improved due to the improvement of the flatness characteristic.

Referring to FIGS. 3A to 3C again, the ends **155**, **165**, and **175** of the driving portion **180** of the resonators **150**, **160**, and **170** (the first end **125** and the second end **145** in FIGS. 1 and 2) may be formed in various shapes.

As exemplarily illustrated in FIG. 3A, the end **155** of the driving portion **180** may be formed such that a width portion of one side of the resonator **150** extends in the length direction thereof, and may be formed in a plate shape up to the thickness of the frame **183**. In the present embodiment, the end **155** may be formed to have a thickness t corresponding to a sum of a thickness t2 of the base **181** and a thickness t1 of the frame **183**, or may be formed to have a thickness greater than the thickness t2 of the base **181** and less than the thickness t.

FIG. 5A is a plan view of an intersecting portion when ends of two facing resonators have plate-shaped ends according to an embodiment. FIG. 5B is a cross-sectional view taken along line A-A* of FIG. 5A according to an embodiment. When the resonator **150** having the end **155** of the driving portion **180** is applied as the first resonator **110** and the second resonator **130** facing each other of the directional acoustic sensor **100** according to the embodiment, as exemplarily illustrated in FIGS. 5A and 5B, the first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** may be arranged in an intersecting structure in which the first end **125** and the second end **145** are engaged with each other. The first end **125** of the first resonator **110** extends in the length direction thereof to correspond to a width portion of one side of the first resonator **110**, and the second end **145** of the second resonator **130** extends in the length direction thereof to correspond to a width portion of the other side of the second

13

resonator 130 that does not face the width portion of one side of the first resonator 110. FIG. 5A is a plan view showing a portion where the first end 125 and the second end 145 intersect each other when the first end 125 of the first resonator 110 and the second end 145 of the second resonator 130 are provided with the end 155 having a plate shape as in FIG. 3A, and FIG. 5B is a cross-sectional view taken along line A-A* of FIG. 5A.

As exemplarily illustrated in FIG. 3B, the end 165 of the driving portion 180 may be formed in a structure in which a width portion of one side of the resonator 160 extends in the length direction thereof, and the frame 183 extends to the end of the extended width portion of the one side.

FIG. 6A is a plan view of an intersecting portion when ends of two resonators facing each other have an end with a frame extending to the end according to an embodiment. FIG. 6B is a cross-sectional view taken along line B-B* of FIG. 6A according to an embodiment. When the resonator 160 having the end 165 of the driving portion 180 is applied as the first resonator 110 and the second resonator 130 facing each other of the directional acoustic sensor 100 according to the embodiment, as exemplarily illustrated in FIGS. 6A and 6B, the first end 125 of the first resonator 110 and the second end 145 of the second resonator 130 may be arranged in an intersecting structure in which the first end 125 and the second end 145 are engaged with each other. The first end 125 of the first resonator 110 may have a structure in which a width portion of one side of the first end 125 of the first resonator 110 extends in the length direction thereof, and the second end 145 of the second resonator 130 may have a structure in which a width portion of other side of the second end 145 of the second resonator 130 that does not face the width portion of the one side of the first resonator 110 extends in the length direction thereof, and frames 127 and 147 may be formed to extend to the end portions of the first and second ends 125 and 145. FIG. 6A is a plan view showing an intersecting portion of the first end 125 and the second end 145 when the first end 125 of the first resonator 110 and the second end 145 of the second resonator 130 facing each other include the end 165 formed by extending the frame 183 to the end as in FIG. 3B, and FIG. 6B is a cross-sectional view taken along line B-B* of FIG. 6A.

As exemplarily illustrated in FIG. 3C, the end 175 of the driving portion 180 may be formed in a structure in which a width portion of one side of the resonator 170 extends in the length direction thereof, and the frame 183 extends to a circumference of the extended end 175.

FIG. 7A is a plan view of an intersecting portion when ends of two resonators facing each other have an end with a frame extending to a circumference thereof according to an embodiment. FIG. 7B is a cross-sectional view taken along line C-C* of FIG. 7A according to an embodiment. When the resonator 170 having the end 175 of the driving portion 180 is applied as the first resonator 110 and the second resonator 130 facing each other of the directional acoustic sensor 100 according to the embodiment, the first end 125 of the first resonator 110 and the second end 145 of the second resonator 130 facing each other may be arranged in an intersecting structure in which the first and second ends 125 and 145 are engaged with each other as exemplarily illustrated in FIGS. 7A and 7B. The first end 125 of the first resonator 110 may have a structure in which a width portion of one side of the first resonator 110 extends in the length direction thereof, and the second end 145 of the second resonator 130 may have a structure in which a width portion of other side of the second resonator 130 that does not face the width portion of one side of the first resonator 110

14

extends in the length direction of the second resonator 130, and the frames 127 and 147 may extend to the circumference of the first end 125 and the second end 145. FIG. 7A is a plan view showing an intersecting portion of the first end 125 and the second end 145 when the first end 125 of the first resonator 110 and the second end 145 of the second resonator 130 facing each other include the end 175 in which the frame 183 is formed by extending to a circumference of the end 175 as shown in FIG. 3C, and FIG. 7B is a cross-sectional view taken along line C-C* of FIG. 7A.

When the resonators 150, 160, and 170 having the ends 155, 165, and 175 of the shape as exemplarily illustrated in FIGS. 3A to 3C are applied to the first resonator 110 and the second resonator 130 facing each other of the directional acoustic sensor 100 according to the embodiment, the intersecting structure in which the first and second ends 125 and 145 are engaged with each other may be formed to a thickness reaching a height of the frame as in FIGS. 5A and 5B, or the frames 127 and 147 may be formed to extend to at least a portion of the engaged intersecting portion of the first and second ends 125 and 145 as in FIGS. 6A and 6B and as in FIGS. 7A and 7B.

FIGS. 1 and 2 illustrate an example in which the structure of FIGS. 5A and 5B is applied as an intersecting structure of the first and second ends 125 and 145 engaged each other. Alternatively, the structure of FIGS. 6A and 6B or the structure of FIGS. 7A and 7B may be applied as the intersecting structure of the first and second ends 125 and 145 engaged each other.

In FIGS. 1, 2, 3A, 3B, and 3C, it is illustrated that the frame 183 does not extend to the end of the base 181 where the first end 125 of the driving portion 120 of the first resonator 110 and the second end 145 of the driving portion 140 of the second resonator 130 are not located, but this is merely an example, and the embodiment is not limited thereto. That is, the frame 183 may extend to the end of the base 181 where the first end 125 and the second end 145 are not located.

In addition, in FIGS. 1, 2, 3A, 3B, and 3C, it has been described and illustrated that the driving portions 120 and 140 of the first resonator 110 and the second resonator 130 have a structure including the base 181 and the frame 183 protruding from the base 181, but this is merely an example, and the embodiment is not limited thereto. For example, the driving portions 120 and 140 of the first resonator 110 and the second resonator 130 may be formed in a structure without the frame 183.

FIG. 8 is a diagram of an example of forming an intersecting structure in which ends of two resonators facing each other are engaged with each other in an edge shape according to an embodiment. FIG. 9 is a diagram of an example of forming an intersecting structure in which ends of two resonators facing each other are engaged with each other in a male/female form according to an embodiment. In the above description, it has been described a case in which, at the intersecting portion of the first end 125 and the second end 145, the width portion of the one side of the first resonator 110 and the width portion of the other side of the second resonator 130 that face each other extend in the length direction thereof to form an intersecting structure in which the first end 125 and the second end 145 are engaged with each other, but the embodiment is not limited thereto. For example, as shown in FIG. 8, the first end 125 and the second end 145 may be formed in an intersecting structure that engages with each other in an edge shape. In addition, an intersecting portion where the first and second ends 125 and 145 are engaged with each other may be formed to have

15

an intersecting structure in which the first end **125** and the second end **145** are engaged with each other in a male/female shape, as shown in FIG. 9. The male/female shape of the first end **125** and the second end **145** may be formed opposite to that in FIG. 9. Also, in FIGS. 8 and 9, the first end **125** and the second end **145** may be formed in a plate-shape with a thickness reaching the height of the frame, or may be formed in a structure in which the frame extends to at least a portion of the intersecting portions of the first and second ends **125** and **145**. In FIGS. 8 and 9, the specific structure of the intersecting portions of the first end **125** and the second end **145** engaged with each other may be inferred from FIGS. 5A and 5B, FIGS. 6A and 6B, and FIGS. 7A and 7B, and thus, repeated descriptions and illustrations thereof are omitted.

In the embodiment described above, as an example, it is depicted that the directional acoustic sensor **100** according to the embodiment has a resonator arrangement including a plurality of first resonators **110**, a plurality of second resonators **130**, and at least one third resonator **130'**, and in FIGS. 1 and 2, the directional acoustic sensor **100** includes a total of 16 resonators by including seven first and second resonators **110** and **130** respectively forming a resonator arrangement to face each other, respectively, and two third resonator **130'** that does not face the first resonator **110**, and the embodiment is not limited thereto.

According to the directional acoustic sensor **100** according to the embodiment as described above, the first end **125** of the first resonator **110** and the second end **145** of the second resonator **130** that face each other in the resonator arrangement form an intersecting structure in which the first end **125** and the second end **145** are engaged with each other, and thus, a through-hole in the cavity may be minimized or virtually eliminated in an arrangement region where the resonators face each other. Accordingly, in the directional acoustic sensor **100** according to the embodiment, a decrease in acoustic resistance may be suppressed, such that sensitivity may be improved, and a layout size of a resonator arrangement may be reduced, such that a device size may be reduced.

On the other hand, when the ends of the resonators facing each other are disposed without an intersecting structure, there may be a through-hole between the two facing resonators, whereby the acoustic resistance is reduced, and as a result, the sensitivity may be reduced than that of the directional acoustic sensor **100** according to the embodiment.

FIG. 10 is a schematic plan view illustrating a directional acoustic sensor according to a comparative example. FIG. 10 shows an example in which the directional acoustic sensor **200** according to the comparative example is provided to compare to the directional acoustic sensor **100** according to the embodiment of FIG. 2.

Referring to FIG. 10, the directional acoustic sensor **200** of the comparative example includes a support **201**, and a plurality of first resonators **210** and a plurality of second resonators **230** provided on the support **201**. The plurality of first resonators **210** and the plurality of second resonators **230** are arranged to form a resonator arrangement in which at least some of them face each other in a cavity of the support **201**.

One end of each of the plurality of first resonators **210** is fixed to the first support portion **203** of the support **201** to form a planarly parallel arrangement in the cavity of the support **201**. One end of each of the plurality of second

16

resonators **230** is fixed to the second support **205** of the support **201** to form a planarly parallel arrangement in the cavity of the support **201**.

When compared with the directional acoustic sensor **100** according to the embodiment shown in FIG. 2, the directional acoustic sensor **200** of the comparative example does not have an intersecting structure of ends of the first resonator **210** and the second resonator **230** facing each other. In FIG. 10, reference numeral **230'** denotes a second resonator that does not face the first resonator **210**.

Each of the plurality of first resonators **210** and the plurality of second resonators **230** include bases **211** and **231** and frames **213** and **233** extending in a length direction thereof to protrude from the bases **211** and **231**, and the frames **213** and **233** extend to end portions of the first resonator **210** and the second resonator **230**.

In the directional acoustic sensor **200** of the comparative example, the first resonator **210** and the second resonator **230** facing each other do not have an intersecting structure of ends, there may be a through hole **240** of the cavity between the first resonator **210** and the second resonator **230** facing each other.

FIG. 11 is a diagram of a comparison of a change of a layout size of a resonator arrangement before (comparative example) and after (embodiment example) application of the intersecting structure of ends of two resonators facing each other according to an embodiment. In FIG. 11, when the resonant frequency change before (comparative example) and after (embodiment example) a change of the shape of the ends of the two facing resonators is limited to about 5% or less, a design example in which the size of device in the length direction thereof is reduced from approximately 2950 μm to approximately 2700 μm is shown. When not limiting the resonant frequency change, the ends that constitute an intersecting structure may be formed up to approximately half the length of the resonator.

In FIG. 11, a layout of the resonator arrangement before applying the intersecting structure of the two facing resonators corresponds to the directional acoustic sensor **200** of the comparative example shown in FIG. 10, and a layout of the resonator arrangement after the application of the intersecting structure of the ends of the two facing resonators corresponds to the directional acoustic sensor **100** according to the embodiment shown in FIG. 2.

As shown in FIG. 11, the directional acoustic sensor **100** according to the embodiment to which an intersecting structure of the ends of the two resonators facing each other is applied has a reduced layout size of the resonator arrangement by about 10% compared to the directional acoustic sensor **200** of the comparative example to which the intersecting structure of the ends is not applied. Therefore, according to the directional acoustic sensor **100** according to the embodiment, the size of the device may be reduced, and thus, price competitiveness may be improved.

FIG. 12 is a graph showing the change in frequency response characteristics of resonators before (comparative example) and after (embodiment example) application of an intersecting structure of ends of two resonators facing each other according to an embodiment.

As shown in FIG. 12, it may be seen that the sensitivity increases by about 2.4 dB when the intersecting structure of the ends of the two resonators facing each other is applied. As such, according to the directional acoustic sensor **100** according to the embodiment, due to the intersecting structure of the ends of the two facing resonators, there is almost no through-hole between the two facing resonators, and

17

thus, the decrease in acoustic resistance is suppressed, thereby increasing the sensitivity.

FIG. 13 is a graph showing resonant frequency characteristics for each channel of a resonator arrangement before (comparative example) and after (embodiment example) the application of an intersecting structure of ends of two resonators facing each other. The channel represents each resonator, and when width, thickness, and mass conditions are the same, the resonant frequency of each channel (resonator) is approximately proportional to the square root of a beam length of the resonator. The resonance frequency graph of FIG. 13 is a result obtained through COMSOL simulation.

As shown in FIG. 13, it may be seen that, even when an intersecting structure of ends of two resonators facing each other is applied, a directional acoustic sensor with almost no change in primary and secondary resonant frequencies may be implemented.

From the difference in layout size of the resonator arrangement before (comparative example) and after (embodiment example) application of the intersecting structure of the ends of the two resonators facing each other as in FIG. 11, the frequency response characteristics of the resonator in FIG. 12, and the resonant frequency characteristics for each channel of FIG. 13, by applying an intersecting structure of ends of two resonators facing each other to a directional acoustic sensor, it may be seen that a directional acoustic sensor having good sensitivity and desired resonant frequency characteristics for each channel may be implemented, and a device size may be reduced. FIGS. 11 to 13 are only illustrative, and the embodiment is not limited thereto.

The directional acoustic sensor 100 according to the embodiment may be manufactured, for example, as shown in FIG. 14, and may be packaged as shown in FIG. 15.

FIG. 14 is a photograph after manufacturing the directional acoustic sensor 100 according to the embodiment. FIG. 15 is a photograph after packaging the directional acoustic sensor 100 of FIG. 14 according to an embodiment. FIG. 15 shows an example in which holes are formed in a packaging case such that acoustic sound is transmitted to the directional acoustic sensor 100. FIG. 16 is a graph showing a result of measuring a frequency response characteristic of the directional acoustic sensor 100 according to the manufactured embodiment. It may be confirmed from the measurement result of FIG. 16 that the device size may be reduced without decreasing in sensitivity. Here, FIGS. 14 to 16 are only illustrative, and the present embodiment is not limited thereto, and the directional acoustic sensor 100 according to the embodiment may be modified to have various frequency response characteristics.

In the above, a resonator having a structure including a base and a frame extending along both edges of the base in the length direction thereof to protrude from the base has been described as the first resonator 110 and the second resonator 130 facing each other in the directional acoustic sensor 100 according to the embodiment, but the embodiment is not limited thereto, and resonators having various structures may be applied.

FIGS. 17, 18 and 19 are diagrams of other examples of resonators that may be applied to two resonators facing each other according to an embodiment. FIGS. 17 to 19 illustrate other examples of resonators 250, 270, and 290 that may be applied as the first resonator 110 and the second resonator 130 facing each other in FIGS. 1 and 2.

Referring to FIG. 17, a driving portion 260 of the resonator 250 may include a base 181 and a frame 183 protrud-

18

ing from the base 181 and extending in the length direction thereof. Also, the driving portion 260 of the resonator 250 may have an end 155 (e.g., corresponding to the first end 125 and the second end 145 in FIGS. 1 and 2) at the end thereof.

The frame 183 may be continuously provided parallel to the length direction of the resonator 250 inside the base 181. In FIG. 17, a case in which only one frame 183 is provided is illustrated as an example, but the embodiment is not limited thereto, and a plurality of frames 183 may be provided inside the base 181.

Referring to FIG. 18, a driving portion 280 of the resonator 270 may include a base 281 and a groove pattern 282 having a predetermined depth in the base 281. Also, the driving portion 280 of the resonator 270 may have an end 155 (corresponding to the first end 125 and the second end 145 in FIGS. 1 and 2) at the end thereof.

The groove pattern 282 may be formed in a regular shape on the base 281. Here, the groove pattern 282 is formed in the base 281 to a predetermined depth, thereby reducing the mass of the resonator compared to a resonator having only the base without the groove pattern.

The resonator 270 according to the present embodiment includes the groove pattern 282 formed to a predetermined depth in the base 281, thereby reducing the mass of the resonator 270 while maintaining the resonant frequency constant, and accordingly, a quality factor may be reduced and a bandwidth may be increased compared to a resonator including only a base without a groove pattern.

Referring to FIG. 19, a driving portion 295 of a resonator 290 may include a base 296 and a plurality of through holes 297 formed through the base 296. The plurality of through-holes 297 may be formed in a regular shape in the base 296. In addition, the driving portion 295 of the resonator 290 may have an end 155 (corresponding to the first end 125 and the second end 145 in FIGS. 1 and 2) at the end thereof.

The resonator 290 shown in FIG. 19 may have the same resonant frequency as a resonator in which a plurality of through holes are not formed. To this end, each through hole 297 may be formed to have a size smaller than the wavelength of an audible frequency band.

In the present embodiment, the resonator 290 includes a plurality of through-holes 297 having a size smaller than a wavelength of an audible frequency band formed in the base 296, thereby reducing the mass of the resonator while maintaining the resonant frequency constant, and accordingly, compared to a resonator that does not include a plurality of through-holes, a quality factor may be reduced and a bandwidth may be increased.

In FIGS. 17, 18, and 19, the resonators 250, 270, and 290 are illustrated as having plate-shaped ends 155 at the ends of the driving portions 260, 280, and 295, but the embodiment is not limited thereto, and the shapes of the ends may be modified in various forms as described above.

On the other hand, the resonators 250, 270, and 290 of FIGS. 17, 18, and 19 may also be applied to the third resonator 130' that does not face the first resonator 110 in FIGS. 1 and 2, and in this case, the structure may be changed to a structure without the end 155, or a structure including the end 155 may be applied as it is.

The directional acoustic sensor 100 according to the embodiment as described above may reduce or eliminate through-holes caused by the arrangement of the resonators having different lengths from each other by forming ends of two facing resonators to an intersecting structure engaged each other, thereby, improving the acoustic sensitivity and reducing the device size.

19

The directional acoustic sensor **100** according to the embodiment described above may be applied to various electronic devices to which a voice interface technology is applied, for example, a smart phone, a foldable phone, an AI speaker, AR glasses, wearable devices, robots, TVs, etc.

FIGS. **20**, **21**, **22**, **23**, **24** and **25** are diagrams of examples of electronic devices to which the directional acoustic sensor according to an embodiment is applied as a voice interface device according to an embodiment. For example, the directional acoustic sensor **100** according to the embodiment may be applied to AR glasses **1000**, earbuds **1100**, smart phones **1200**, TVs **1300**, artificial intelligent speakers **1400**, and the like as shown in FIGS. **20** to **24**. Also, the directional acoustic sensor **100** according to the embodiment may be applied to a vehicle **1500** as shown in FIG. **25**. FIG. **25** shows an example in which the directional acoustic sensor **1510** according to the embodiment is provided on an upper side of the windshield, and besides above, the directional acoustic sensor **1510** according to the embodiment may be mounted at various positions inside and outside the vehicle **1500**. In addition, the directional acoustic sensor **100** according to the embodiment may be applied to various electronic devices including household appliances, such as air conditioners, refrigerators, and air purifiers. Although the embodiments have been described above, this is merely examples, and various modifications are possible to those skilled in the art from the above description.

According to the directional acoustic sensor according to the embodiment, by forming ends of two facing resonators in an intersecting structure, a through-hole of a cavity in an arrangement region facing resonators may be minimized, and thus a directional acoustic sensor with improved sensitivity may be realized.

In addition, by forming the ends of the two resonators in an intersecting structure, the size of the resonator arrangement region may be reduced, thereby reducing the device size.

It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

1. A directional acoustic sensor comprising:

a support comprising a first support portion and a second support portion that are separated from each other and face each other;

a plurality of first resonators extending in a length direction thereof from the first support portion of the support, each first resonator of the plurality of first resonators comprising:

a first end fixed to the first support portion and having a first width at a position where the first end is fixed to the first support portion;

a base and a frame extending from the first end, wherein the frame extends to a first height and the base has a thickness that is less than the first height; and

a second end opposite the first end that is fixed to the first support portion and having a second width less than the first width; and

20

a plurality of second resonators extending in the length direction thereof from the second support portion of the support and facing the plurality of first resonators, each second resonator of the plurality of second resonators comprising:

a first end fixed to the second support portion and having a third width at a position where the first end is fixed to the second support portion;

a base and a frame extending from the first end, wherein the frame extends to the first height and the base has a thickness that is less than the first height; and

a second end opposite the first end that is fixed to the second support portion and having a fourth width less than the third width,

wherein, in a first resonator arrangement of a region where the plurality of first resonators and the plurality of second resonators face each other, the second ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.

2. The directional acoustic sensor of claim 1, wherein the second end of each first resonator of the plurality of first resonators has a structure in which a width portion of each first resonator extends in the length direction thereof, and wherein the second end of each second resonator of the plurality of second resonators has a structure in which a width portion of each second resonator that does not face the width portion of a corresponding first resonator extends in the length direction thereof.

3. The directional acoustic sensor of claim 2, wherein the second ends of the plurality of first resonators and the second ends of the plurality of second resonators respectively engage with each other.

4. The directional acoustic sensor of claim 2, wherein the second ends of the plurality of first resonators and the second ends of the plurality of second resonators respectively engage with each other in a male or female form.

5. The directional acoustic sensor of claim 1, wherein, in each first resonator of the plurality of first resonators and each second resonator in the plurality of second resonators, the frame is integrally formed with the base.

6. The directional acoustic sensor of claim 1, wherein, in each first resonator of the plurality of first resonators and each second resonator in the plurality of second resonators, each of the base and the frame comprises silicon.

7. The directional acoustic sensor of claim 1, wherein, in each first resonator of the plurality of first resonators and each second resonator in the plurality of second resonators, the frame is provided on at least one of both sides and an inside of the base.

8. The directional acoustic sensor of claim 1, wherein each the second ends of the plurality of first resonators and the second ends of the plurality of second resonators has a plate shape up to the first height.

9. The directional acoustic sensor of claim 1, wherein, in each first resonator of the plurality of first resonators and each second resonator in the plurality of second resonators, the base and the frame respectively extend to each of the second ends of the plurality of first resonators and each of the second ends of the plurality of second resonators.

10. The directional acoustic sensor of claim 1 wherein the frame of each first resonator and the frame of each second resonator extends to end portions of the second end of each first resonator and the second end of each second resonator, respectively.

21

11. The directional acoustic sensor of claim 1, wherein the frame of each first resonator and the frame of each second resonator extends to a circumference of the second end of each first resonator and the second end of each second resonator, respectively.

12. The directional acoustic sensor of claim 1, further comprising at least one third resonator extending in the length direction thereof from the first support portion or the second support portion of the support and not facing the plurality of first resonators or the plurality of second resonators.

13. The directional acoustic sensor of claim 12, wherein the at least one third resonator comprises:

- a base; and
- a frame protruding from the base and extending in the length direction thereof.

14. The directional acoustic sensor of claim 1, wherein each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators comprises:

- a driving portion configured to move in response to an input sound signal; and
- a sensing portion configured to detect movement of the driving portion.

15. The directional acoustic sensor of claim 14, wherein the second ends of each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators are configured to move freely.

16. The directional acoustic sensor of claim 14, wherein the plurality of first resonators and the plurality of second resonators have resonant frequencies different from each other.

17. An electronic device comprising:

a directional acoustic sensor comprising:

- a support comprising a first support portion and a second support portion that are separated from each other and face each other;

a plurality of first resonators extending in a length direction thereof from the first support portion of the support, each first resonator of the plurality of first resonators comprising:

- a first end fixed to the first support portion and having a first width at a position where the first end is fixed to the first support portion;

a base and a frame extending from the first end, wherein the frame extends to a first height and the base has a thickness that is less than the first height; and

a second end opposite the first end that is fixed to the first support portion and having a second width less than the first width; and

a plurality of second resonators extending in the length direction thereof from the second support portion of the support and facing the plurality of first resonators, each second resonator of the plurality of second resonators comprising:

- a first end fixed to the second support portion and having a third width at a position where the first end is fixed to the second support portion;

a base and a frame extending from the first end, wherein the frame extends to the first height and the base has a thickness that is less than the first height; and

a second end opposite the first end that is fixed to the second support portion and having a fourth width less than the third width,

22

wherein, in a first resonator arrangement of a region where the plurality of first resonators and the plurality of second resonators face each other, the second ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.

18. The electronic device of claim 17, wherein the directional acoustic sensor further comprises at least one third resonator extending in the length direction thereof from the first support portion or the second support portion of the support and not facing the plurality of first resonators or the plurality of second resonators.

19. The electronic device of claim 18, wherein the at least one third resonator comprises:

- a base; and
- a frame protruding from the base and extending in the length direction thereof.

20. The electronic device of claim 17, wherein each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators comprises:

- a driving portion configured to move in response to an input sound signal, and
- a sensing portion configured to detect movement of the driving portion.

21. The electronic device of claim 20, wherein the second ends of each first resonator of the plurality of first resonators and each second resonator of the plurality of second resonators are configured to move freely.

22. The electronic device of claim 20, wherein the plurality of first resonators and the plurality of second resonators have resonant frequencies different from each other.

23. A directional acoustic sensor comprising:

a support;

a plurality of first resonators extending in a length direction thereof from the support, each first resonator of the plurality of first resonators comprising:

a first end fixed to the support and having a first width at a position where the first end is fixed to the support;

a base and a frame extending from the first end, wherein the frame extends to a first height and the base has a thickness that is less than the first height; and

a second end opposite the first end that is fixed to the support and having a second width less than the first width; and

a plurality of second resonators extending in the length direction thereof from the support and facing the plurality of first resonators, each second resonator of the plurality of second resonators comprising:

a first end fixed to the support and having a third width at a position where the first end is fixed to the support;

a base and a frame extending from the first end, wherein the frame extends to the first height and the base has a thickness that is less than the first height; and

a second end opposite to the first end that is fixed to the support and having a fourth width less than the third width,

wherein the second ends of the plurality of first resonators and the second ends of the plurality of second resonators form an intersecting structure.